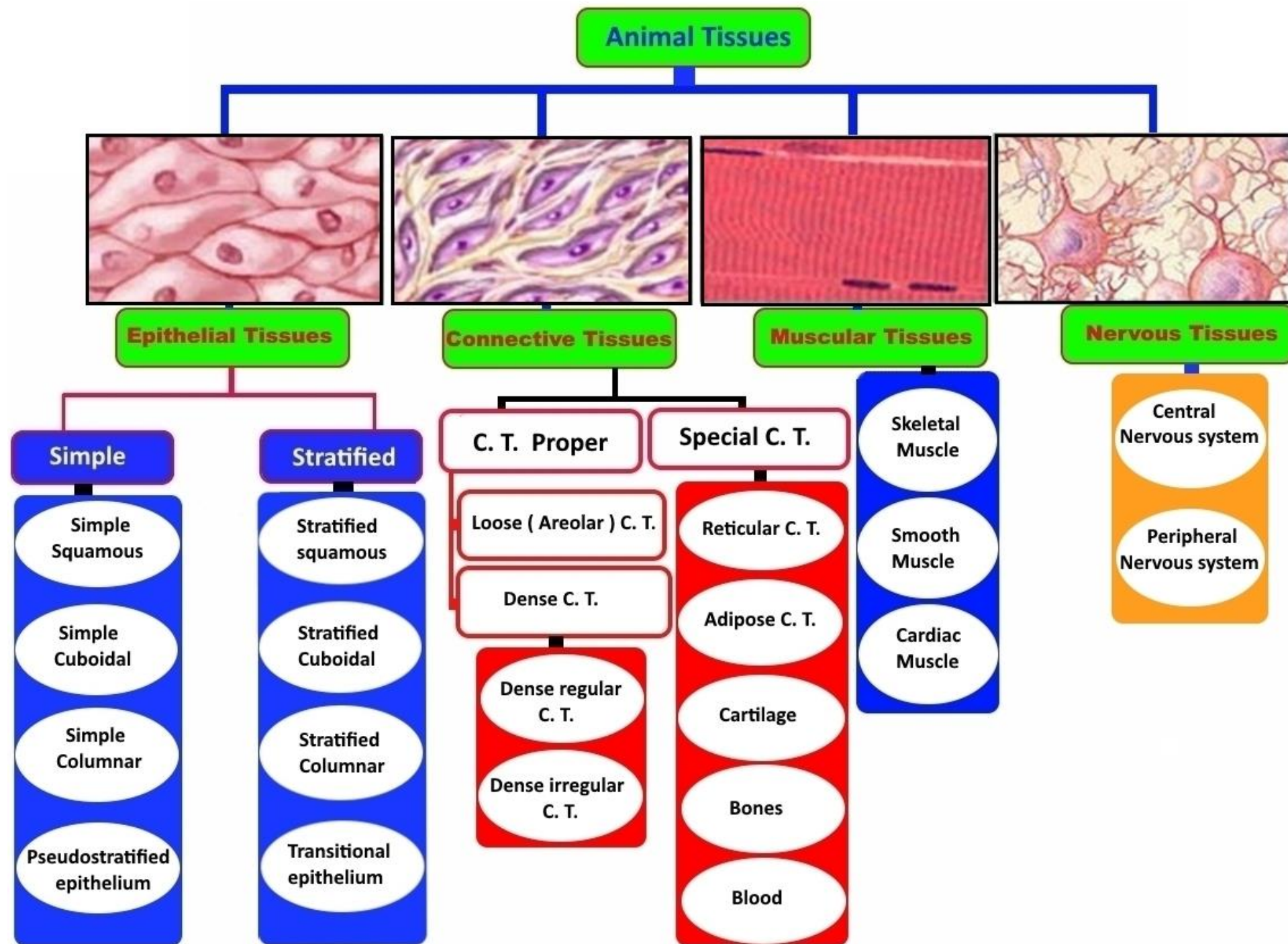


Animal Tissues

Animal Tissues

- **Tissues** are a group of similar cells from the exact origin that together carries out a specific function, it is a cellular organizational level intermediate between cells and a complete organs.
- **Histology is the study of animal tissues**
- **There are four types tissues in the animals:**
 1. **Epithelial tissues**
 2. **Connective tissues**
 3. **Muscle tissues**
 4. **Nervous tissues**
- All animals and humans contain the four types of tissues but particular tissue may differ developmentally for a different classification of animals.



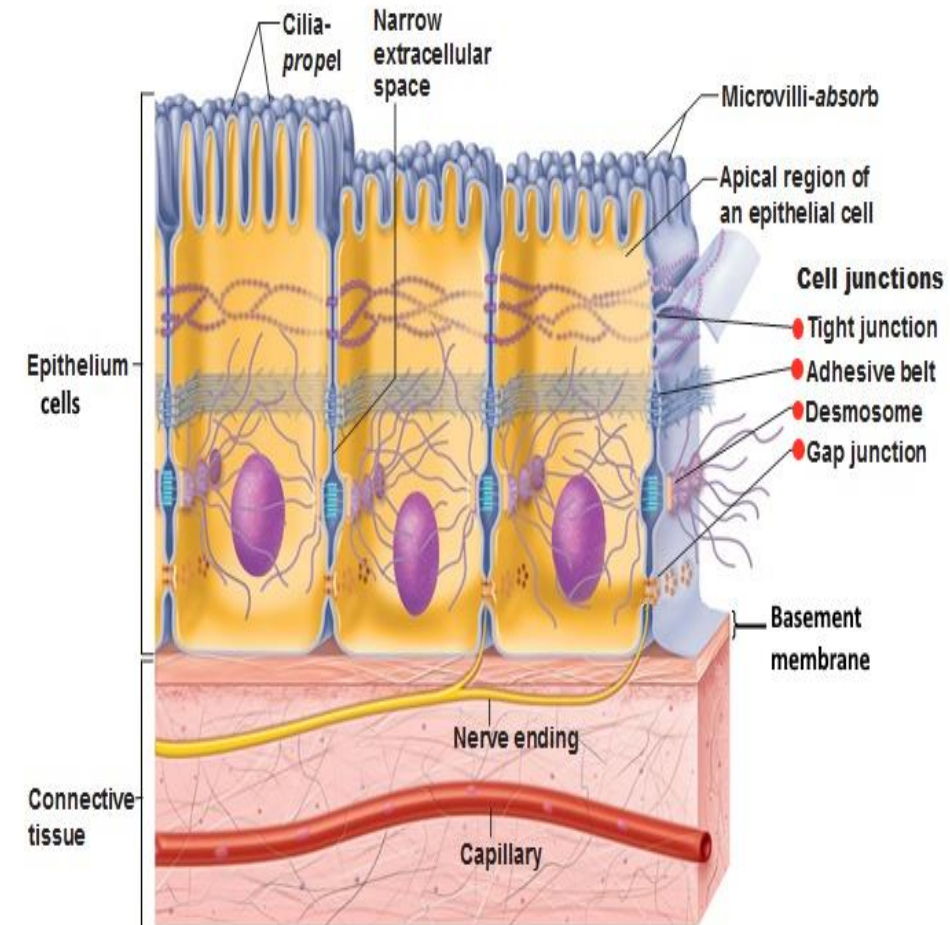
The Fundamental Aims of Studying Histology

1. Understanding tissue structure at levels not visible to the unaided eye.
2. Understanding the relationship between tissue structure and function.
3. Providing a basis for learning histopathology (The relationship between abnormal tissue structure and functional defects).
4. Providing a basis for treating diseased and injured tissues.

Epithelial Tissues

Epithelial Tissues

- True Epithelium compose of cells held together by tight junctions.
- Separated from other below tissue by a basal membrane (basement membrane)
- Non-vascularized (**No blood vessels**)
- **Nourishment:** by diffusion of substances from the underlying connective through the basement membrane
- Cover all organismal surfaces that are in contact with the external environments such as skin, airways, and digestive tracts



Epithelial Tissues Functions:

1. Protection of underlying tissues
2. Diffusion (movement of molecules down their concentration gradients)
3. Absorption (taking molecules into cells)
4. Secretion (releasing molecules from cells)
5. Ion transport (moving charged ions across the tissue)
6. Filtration (removal of molecules from fluid)
7. Forms slippery surfaces (production of mucus)

Epithelial tissues divide according to

- ✓ the shape of epithelial cells
- ✓ and the number of layers

- **One layer:**

1. Simple squamous
2. Simple cuboidal
3. Simple columnar
4. Pseudostratified epithelium

- **More than one layer:**

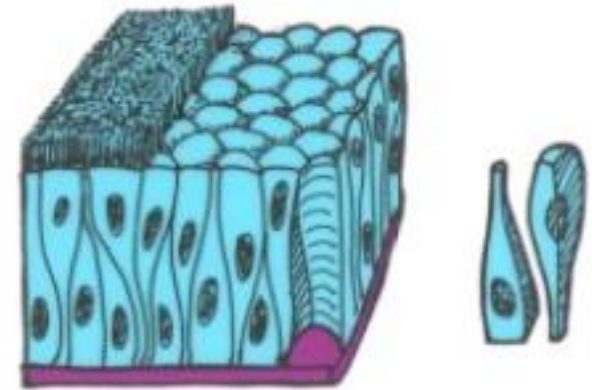
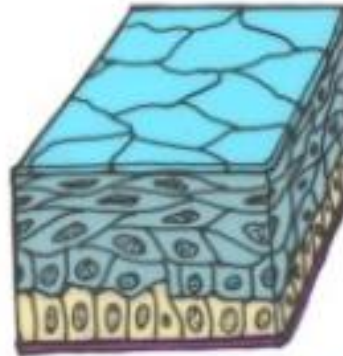
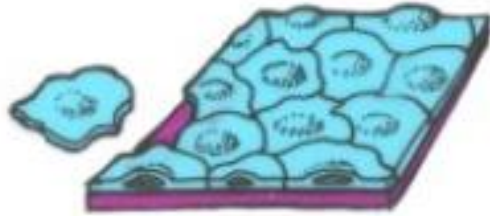
1. Stratified squamous
2. Stratified cuboidal
3. Stratified columnar
4. Transitional epithelium

Simple

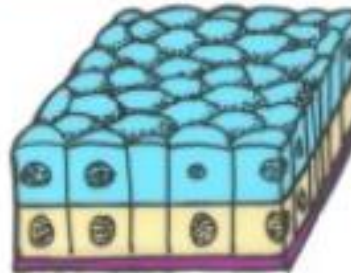
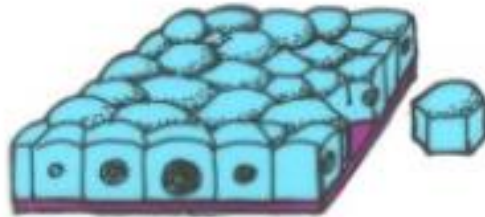
Stratified

Pseudostratified Columnar

Squamous

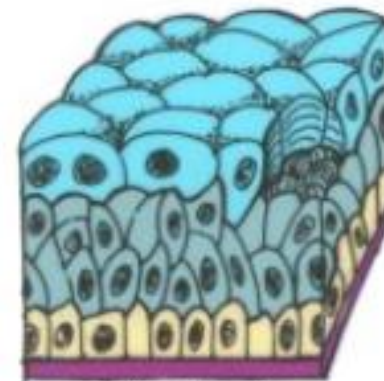
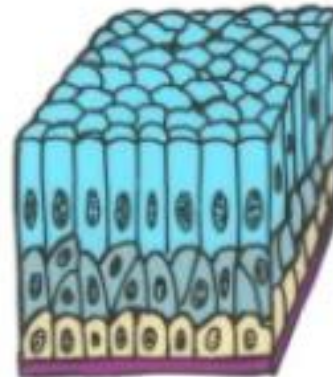
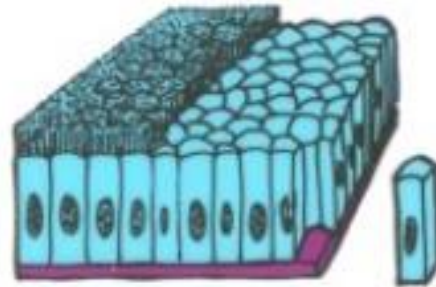


Cuboidal



Transitional

Columnar

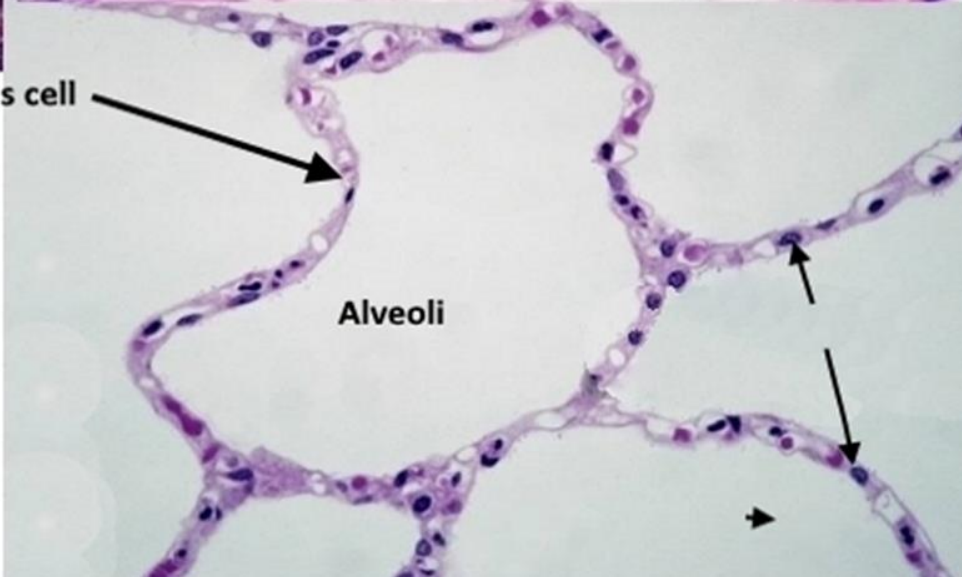
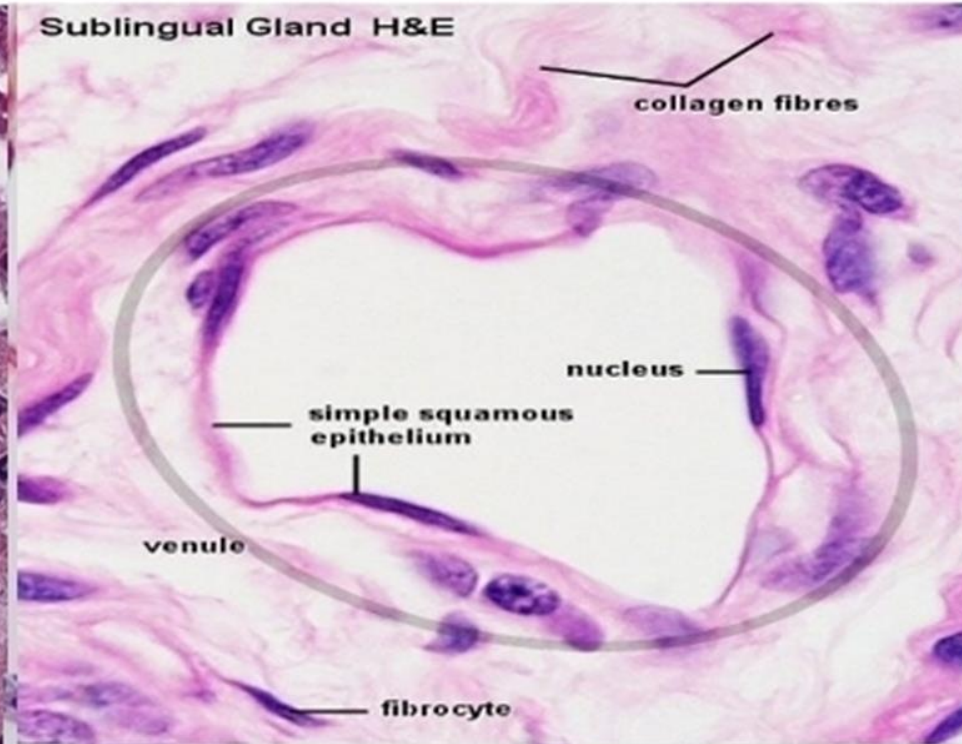
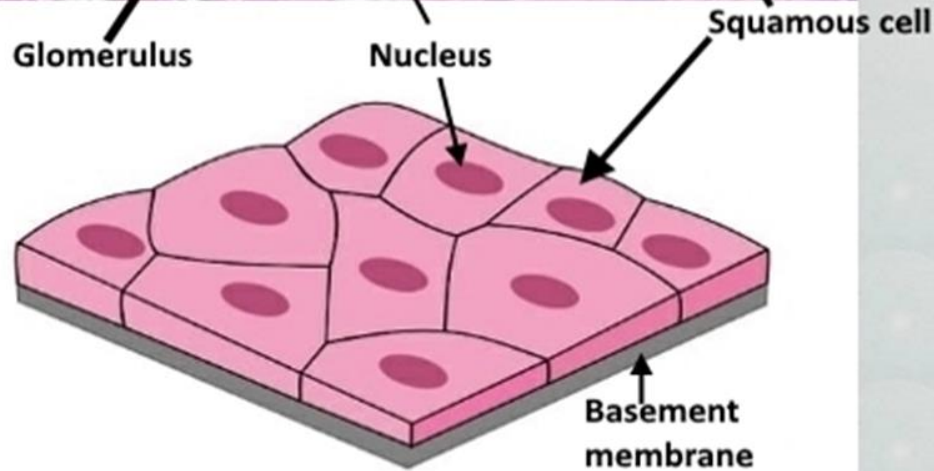


Simple Epithelial Tissue

Simple Squamous Epithelial Tissue

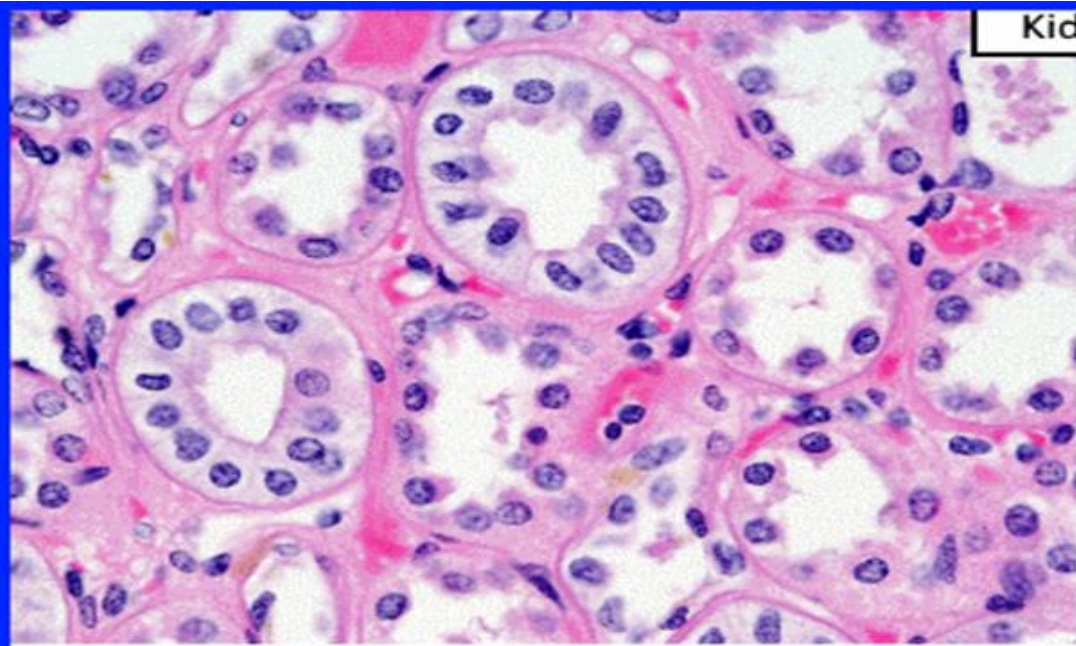
- **Squamous epithelial cells** appear thin, scale-like, flattened, or oval-like shape
- They provide a smooth, low-friction surface over which fluids can move easily
- Functions:
 - ✓ Allow materials to pass through by diffusion or filtration
 - ✓ Secrete lubricating materials
- Found in:
 - ✓ Surface of the skin
 - ✓ Lining surfaces utilizing simple passive diffusion such as the alveolar epithelium in the lungs and lining the walls of capillaries
 - ✓ Lining of the pericardial

Simple Squamous

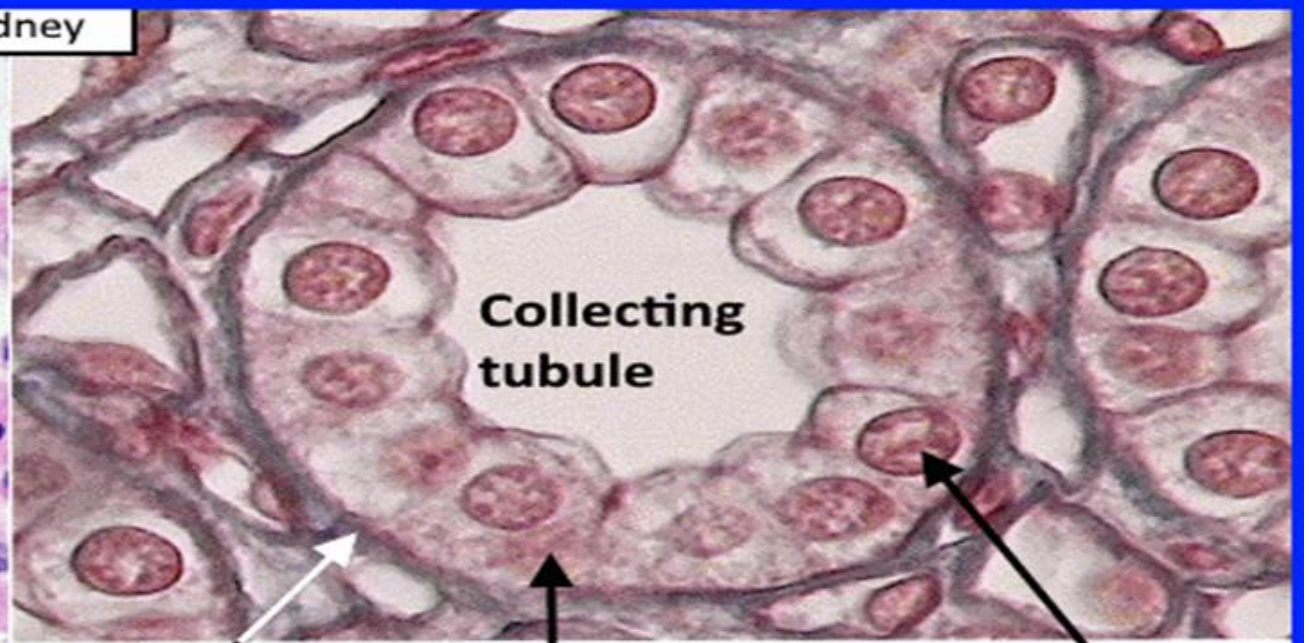


Simple Cuboidal Epithelial Tissue

- Cells are roughly cuboidal in shape, appearing square in cross section
- The cell has a spherical nucleus in the center
- Function: secretion, absorption, or excretion
- These tissues are found in secretive tissue (exocrine gland such as pancreas) or absorptive tissue (lining of the kidney duct and tubules as well as in the ducts of the glands and salivary gland)



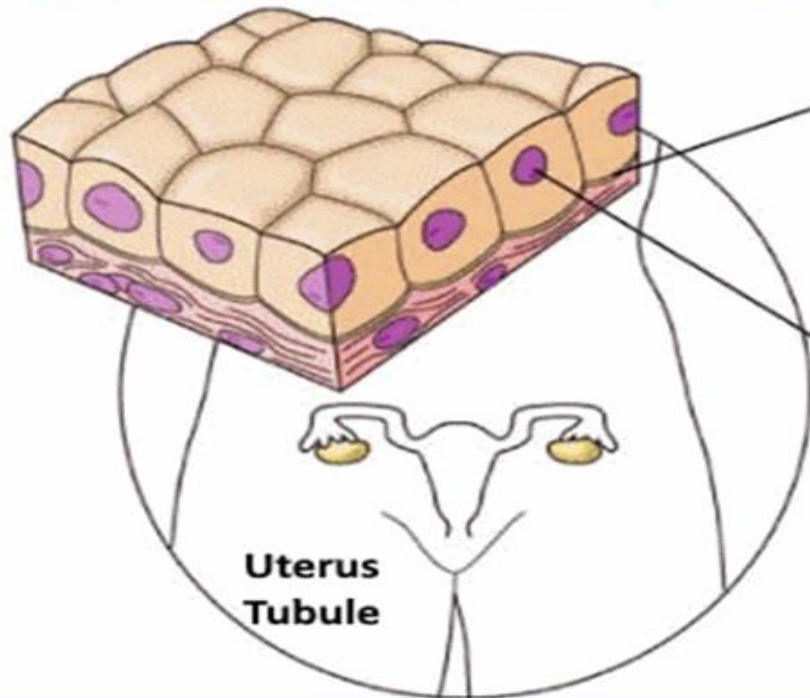
Kidney



Collecting tubule

Simple Cuboidal

Nucleus



Uterus
Tubule

Basement
membrane

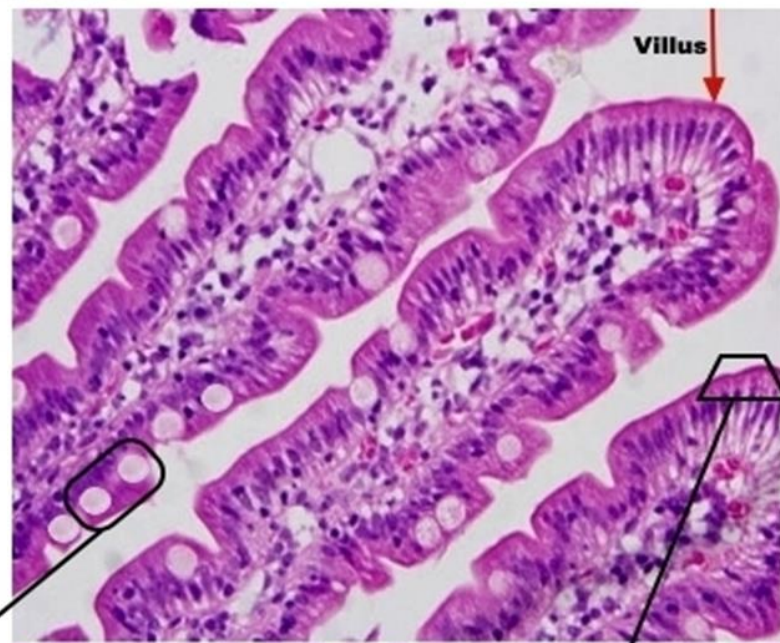
Nucleus of
simple
cuboidal
epithelial
cell

Simple cuboidal

Simple Columnar Epithelial Tissue

- Cells are elongated and column-shaped
- Their nuclei are elongated and are usually located near the base of the cells
- Simple columnar possess cellular extensions; either microvilli (as in the small intestine), or cilia (as the female reproductive tract)
- **Function:** Secretion (as in wall of the stomach), Absorption (as in small intestine) , and excretion.
- The columnar epithelial may has Goblet cells
 - ✓ Goblet cells: a unicellular glands that are found between the columnar epithelial cells secrete mucus, which acts as a lubricant

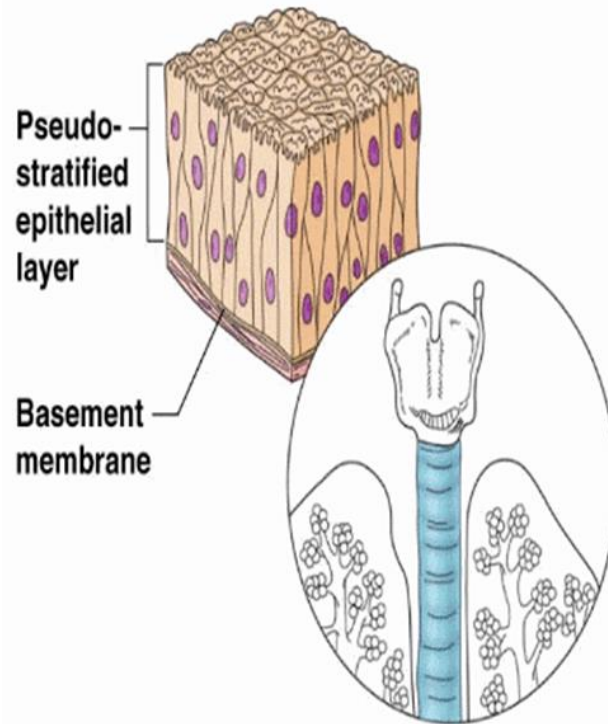
Simple Columnar in intestine



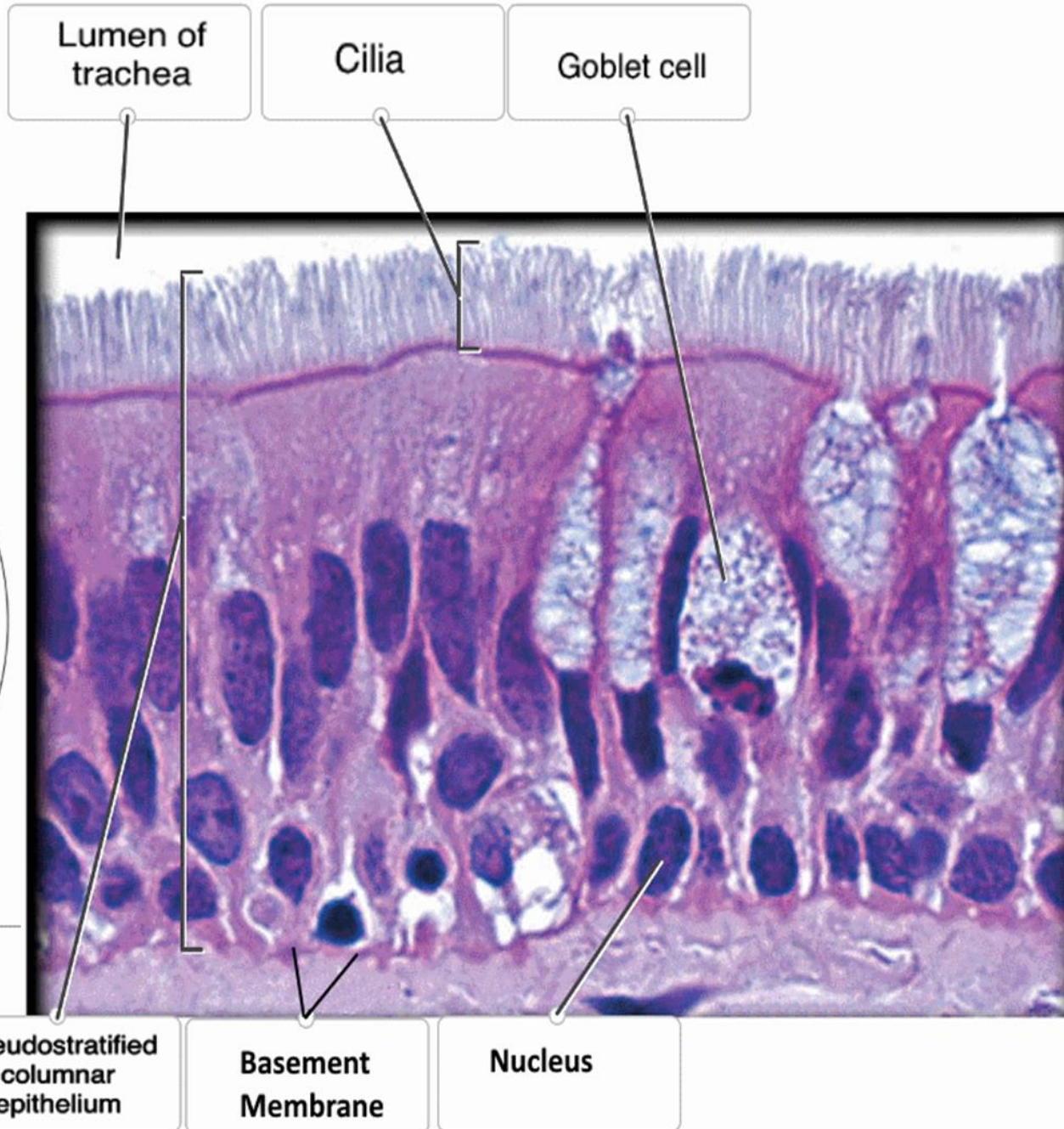
Pseudostratified Epithelial Tissue

- The cells are simple columnar epithelial cells
- The cells have nuclei appear at different heights, giving the misleading impression that the epithelium is stratified when the cells are viewed in cross section
- All cells are attached to the basement membrane
- The tissue may be ciliated or non-ciliated. The ciliated type is also called respiratory epithelium
- The Goblet cells secrete Mucus to lubricate and to trap pathogens and particles
- The ciliated Pseudostratified epithelium found in:
 1. Large respiratory airways of the nasal cavity, trachea and bronchi where the Cilia sweep the trapped things in mucus away from the lung.
 2. The uterus and Fallopian tubes of females, where the cilia propel the ovum to the uterus.

Pseudostratified Epithelium



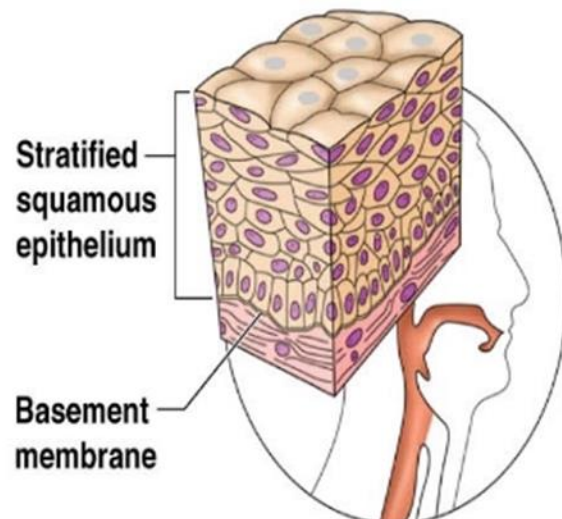
(d) Diagram: Pseudostratified (ciliated) columnar



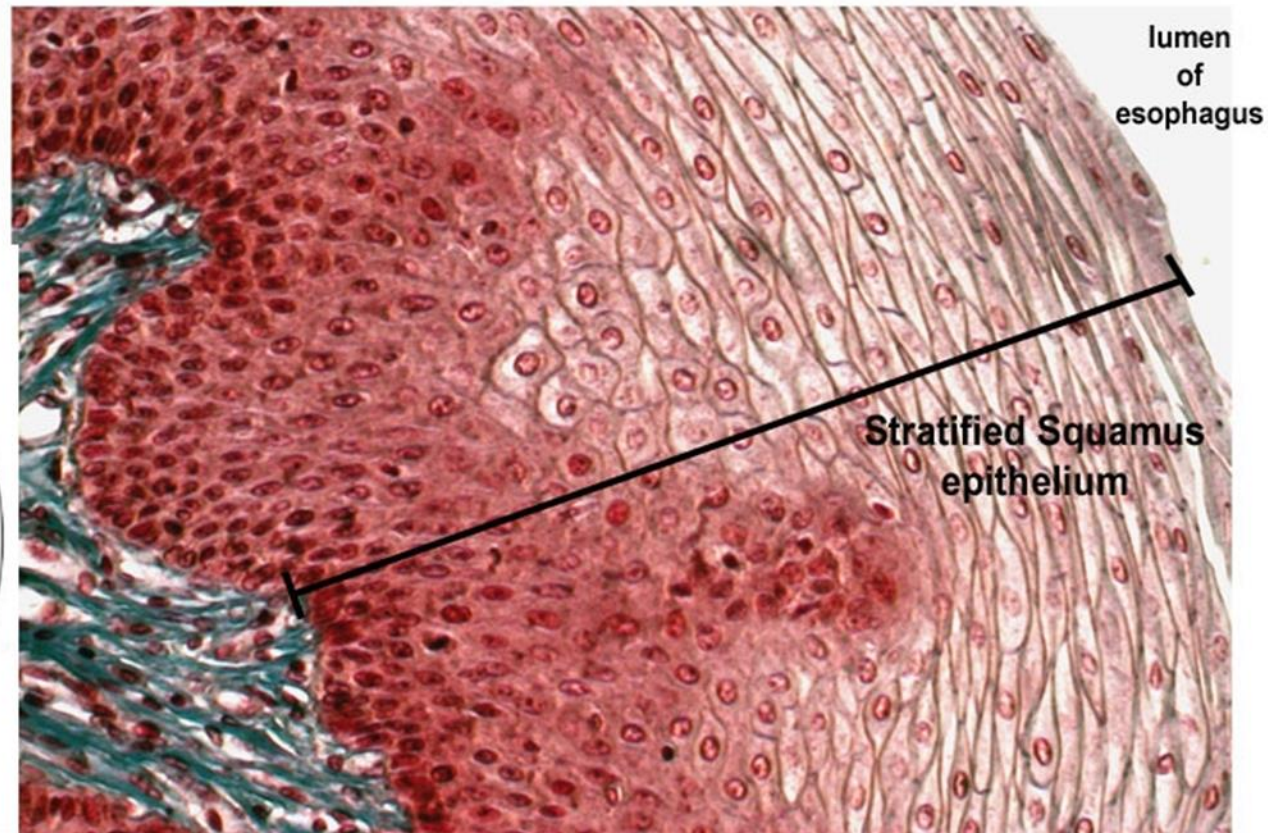
Stratified Epithelial Tissue

Stratified Squamous Epithelial Tissue

- Several layers of cells; cuboidal to columnar shape in deep layers and squamous in apical layer
- Lining of the mouth, esophagus, part of epiglottis, vagina, and covers the tongue

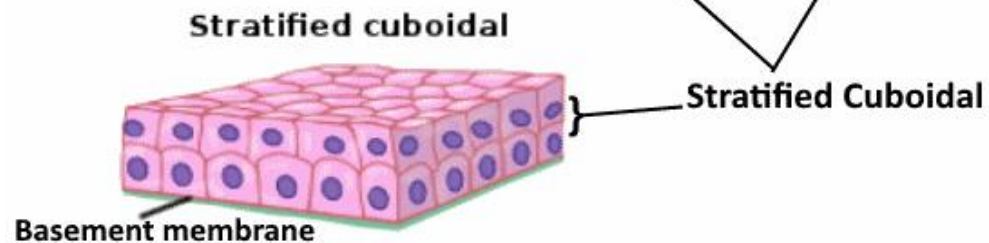
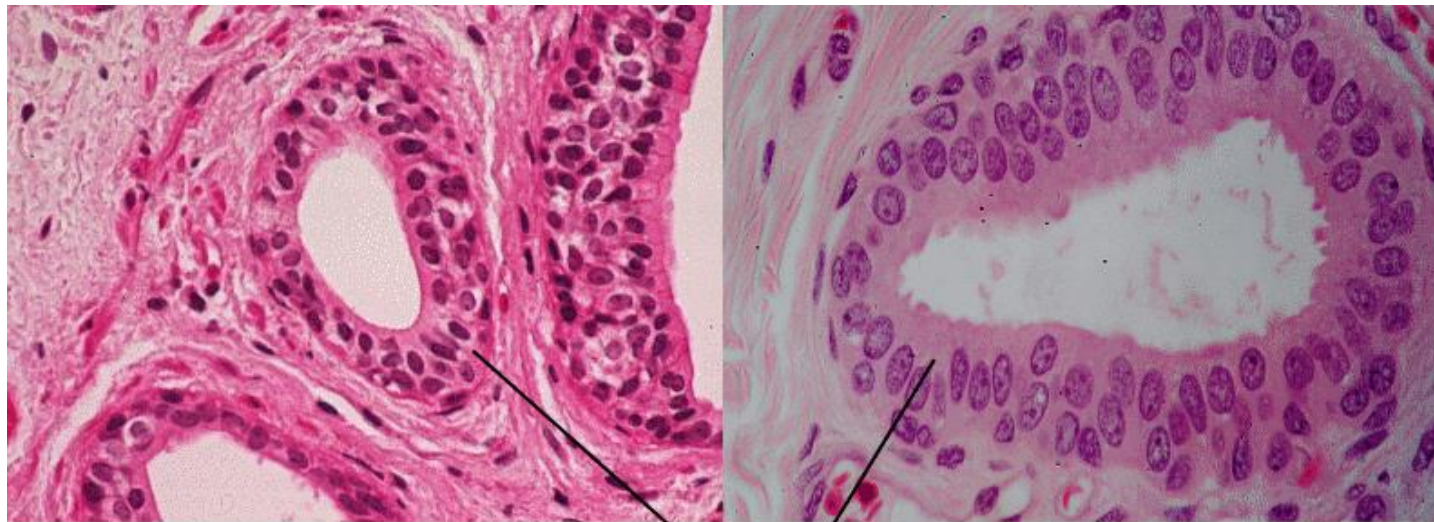


(e) Diagram: Stratified squamous



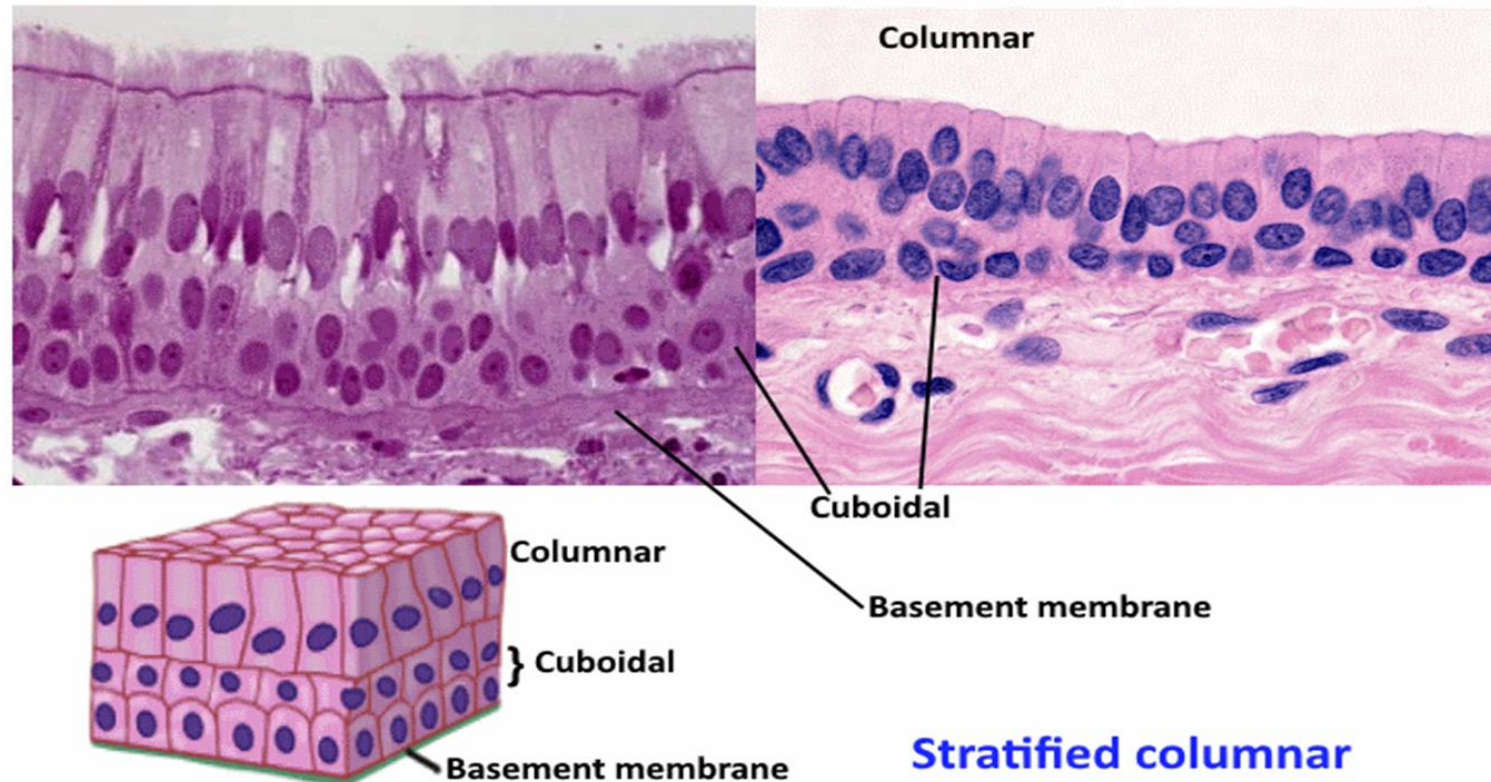
Stratified Cuboidal Epithelial Tissue

- Two or more layers of cells in which cells in the apical are cube-shaped
- Functions in protections, limited secretion and absorption
- Located in ducts of adult sweat and esophageal glands, and parts of male urethra.



Stratified Cuboidal Epithelial Tissue

- Basal cells usually cuboidal; superficial cells elongated and columnar
- It functions in protection and secretion
- Stratified columnar is rare in the body; small amounts in male urethra and in large ducts of some glands

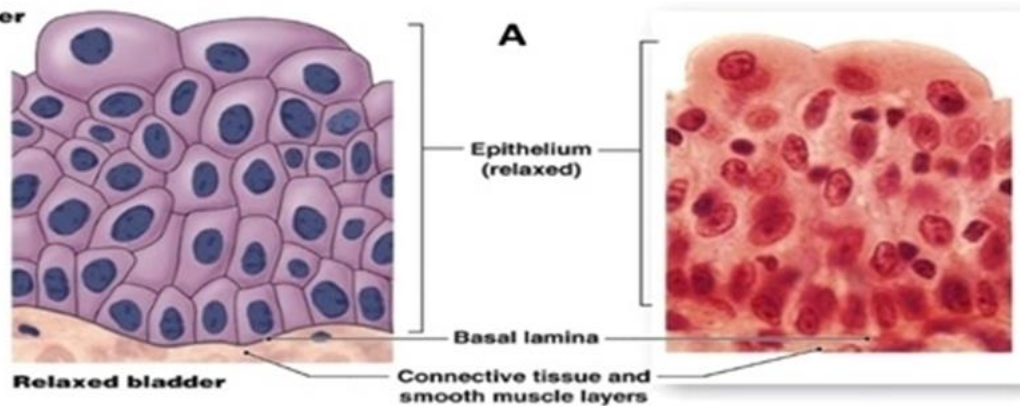


Transitional Epithelial Tissue

- Transitional epithelia are found in tissues that stretch appear to be stratified cuboidal when the tissue is relaxed, or stratified squamous when the organ is distended and the tissue stretches
- It is almost exclusively found in the bladder, ureters and urethra

Epithelium in a Relaxed Bladder

In an empty urinary bladder, the superficial cells are cuboidal with a dome-shaped surface.



Epithelium in a Stretched Bladder

When the urinary bladder is full, the volume of urine has stretched the lining to such a degree that the epithelium appears flattened, and more like a stratified squamous epithelium.

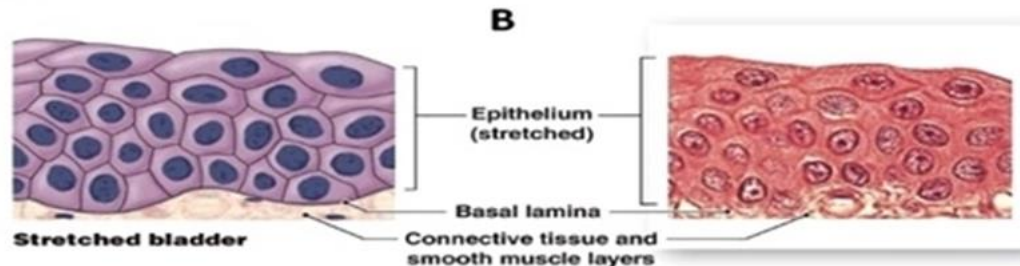
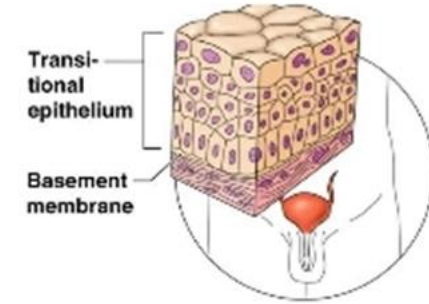
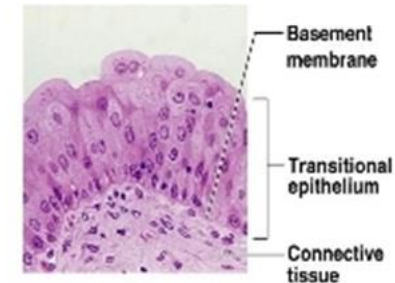


Figure 11.5: Transitional epithelium: A) Relaxed; B) Stretched



(f) Diagram: Transitional



Photomicrograph: Transitional epithelium lining of the bladder, relaxed state (300x); note the rounded appearance of the cells at the surface; these cells flatten and elongate when the bladder fills with urine.

Connective Tissues

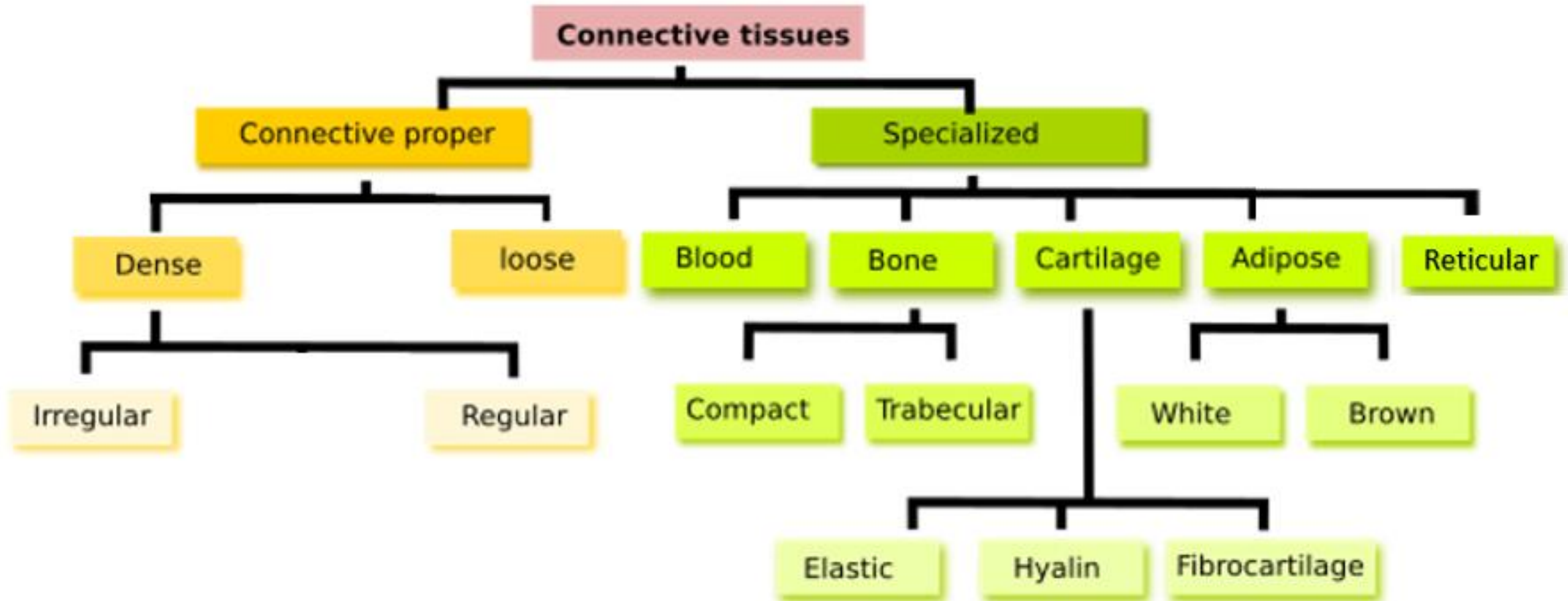
Connective Tissues

- **Connective tissues** are fibrous tissues, made up of cells separated by extracellular matrix (non-living material)
- All connective tissue (except blood and lymph) consists of three main components
 - ✓ Fibers
 - ✓ Ground substance
 - ✓ Cells

Connective tissues contain three types of fibres:

1. **Collagen fibers** are the most widespread and made up of fibrous protein, collagen. Collagen fibres are flexible and have high tensile strength (comparable to steel).
 2. **Elastic fibers** form a network and can be stretched like a rubber band. They are made up of protein elastin. They retain their original shape and size once the force is removed.
 3. **Reticulate fibers** consist of collagen and glycoproteins. They are thin and form a delicate network. They join connective tissues to neighbouring tissues.
- Blood and lymph lack the fiber components
 - It supports, connects, or separates different types of tissues and organs in the body

Connective Tissue Types



Connective tissue proper

- **Connective tissue proper includes:**
 1. **Loose (Areolar) Connective Tissue**
 2. **Dense Connective Tissue**
 - ✓ **Dense Irregular Connective Tissue**
 - ✓ **Dense Regular Connective Tissue**

Loose (Areolar) Connective Tissue

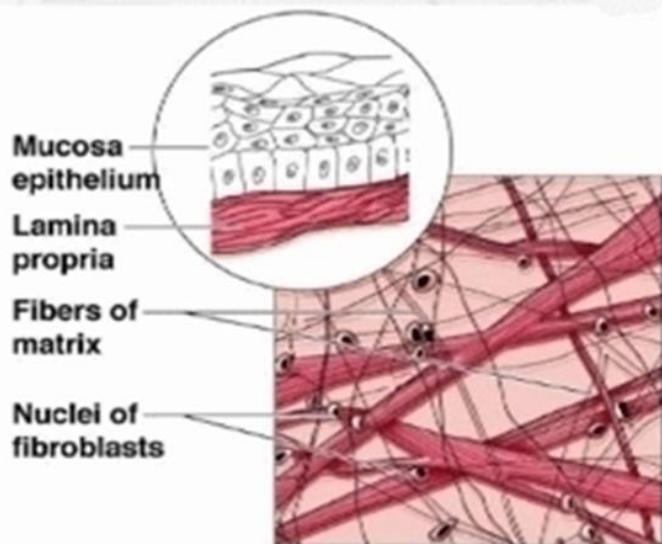
- Most common type of connective tissue in vertebrates
- Strong enough to bind different tissues together and soft enough to provide flexibility and cushioning
- **Function:** holds organs in place and attaches epithelial tissue to underlying tissue
- **Location:** surrounds blood vessels and nerves
- The matrix of this tissue contains many different cells such as **mast cells and microphages.**
- The matrix contains three main types of connective tissue fiber, **collagen fibers, elastic fibers, and reticular fibers**

Loose (Areolar) connective tissue

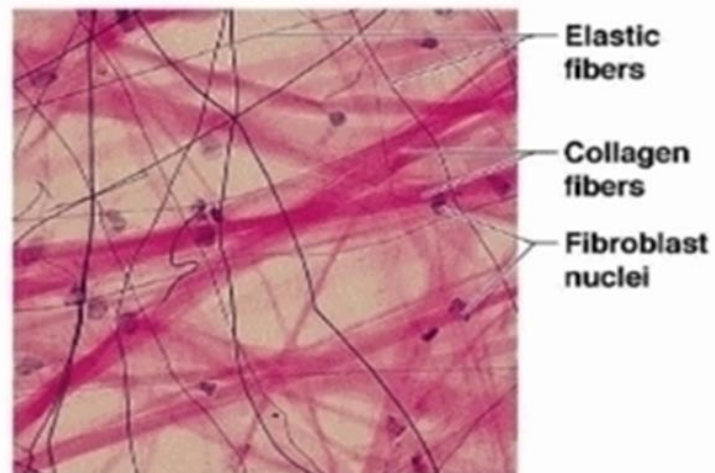
Elastin Fiber
Fibroblast
Collagen fiber



Reticular fiber
Microphage
Plasma cell
Adipocyte (fat cell)
Ground substance
Mast Cells
Elastin fibers
Collagen fibers
Fibroblast
Macrophages
Mesenchymal cell
Lymphocytes
Blood vessel



(e) Diagram: Areolar



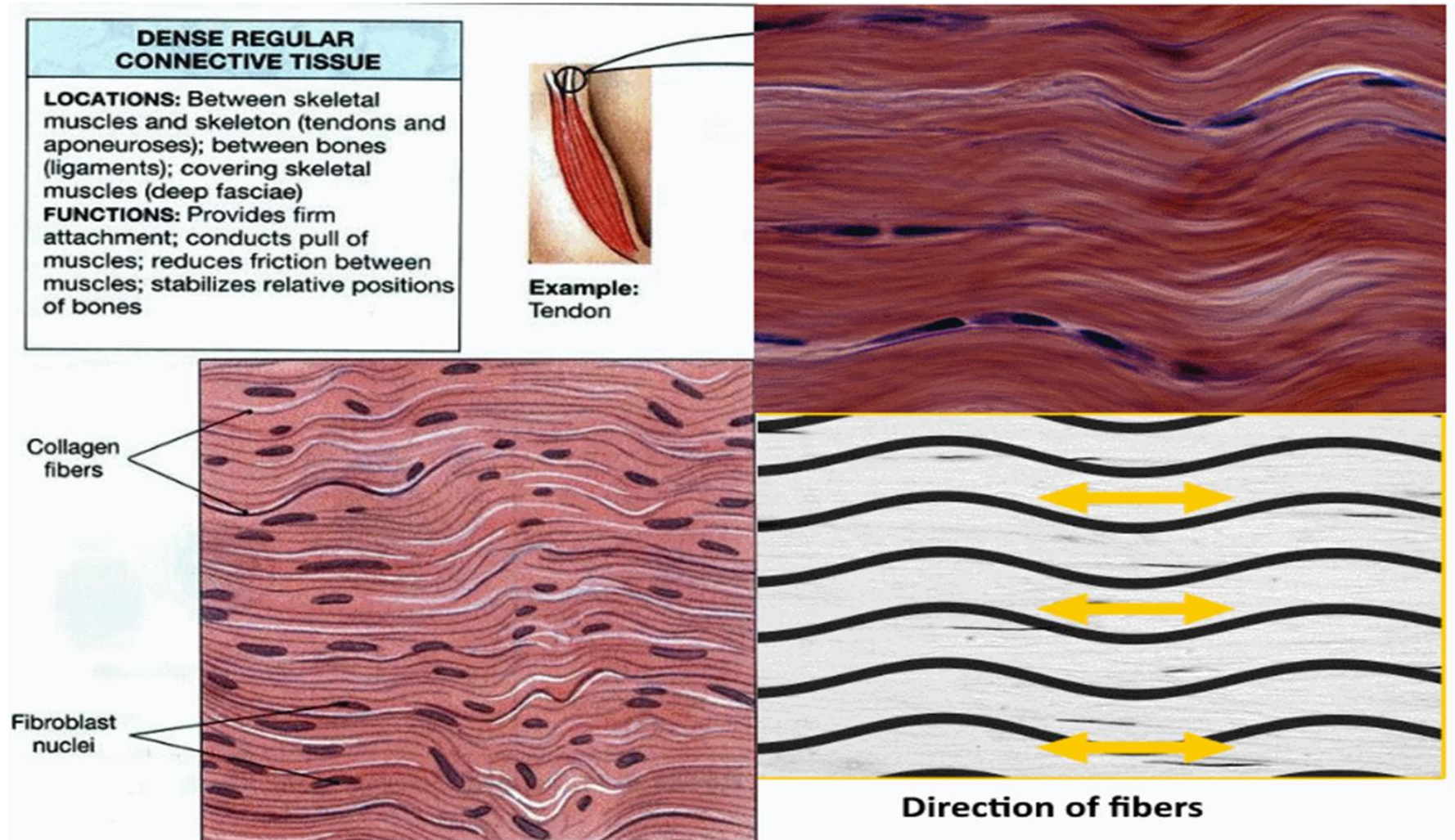
Photomicrograph: Areolar connective tissue, a soft packaging tissue of the body

Dense Connective Tissue

- **Extracellular matrix:** collagen fibers
- **Cell types:** rows of fibroblasts
- Dense connective tissue forms strong, rope-like structures (such as tendons and ligaments} or arranged in sheets (as in the lower layers of the skin (dermis))
- **There are two types of dense connective tissue:**
 1. Dense irregular connective tissue
 2. Dense regular connective tissue

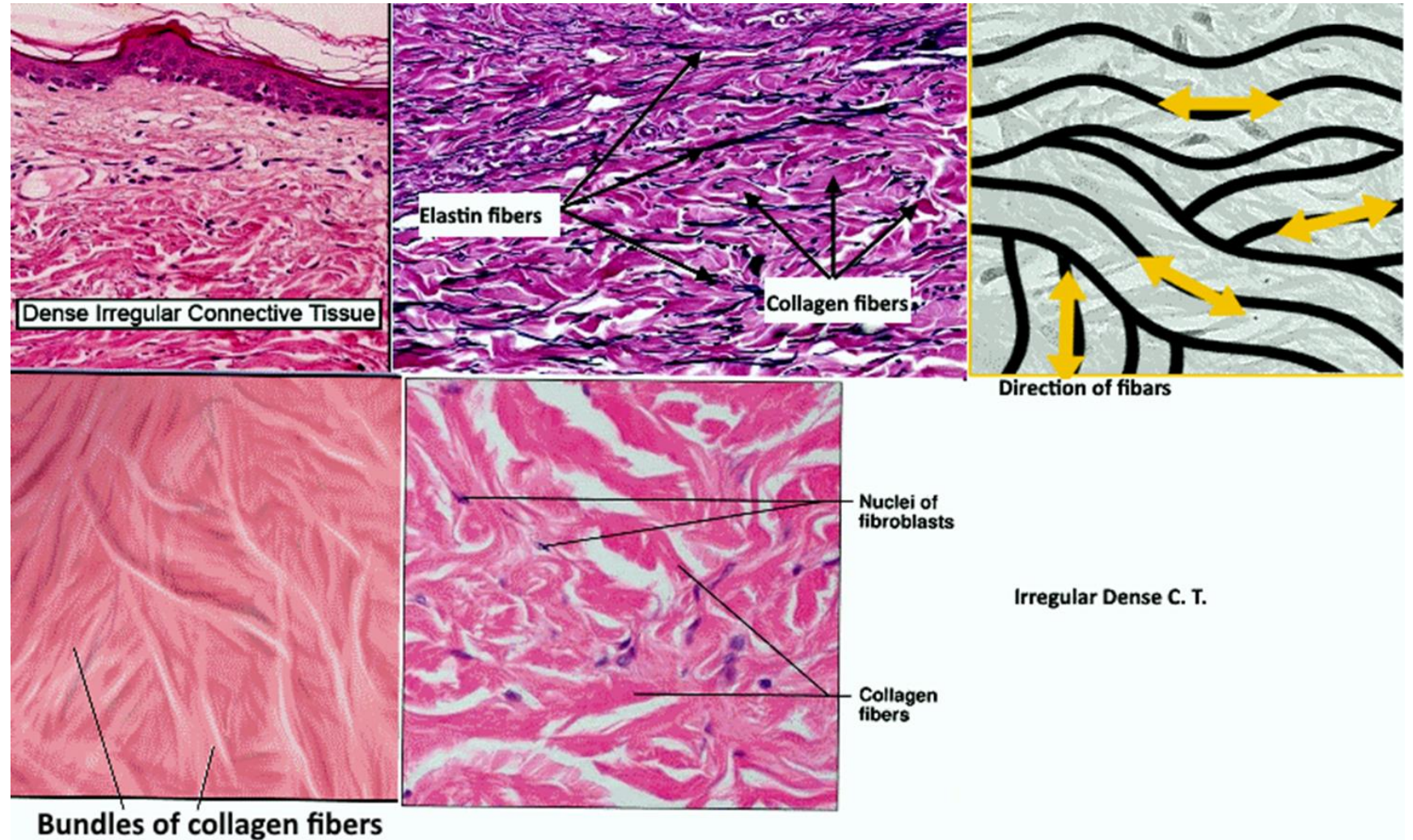
1. Dense Regular Connective Tissue

- Collagen fibers are bundled in a parallel fashion
- Elastic and reticular fibers are completely absent



2. Dense Irregular Connective Tissue

- Irregular patterns/network of collagen bundles and matrix and sometimes found in the form of a network



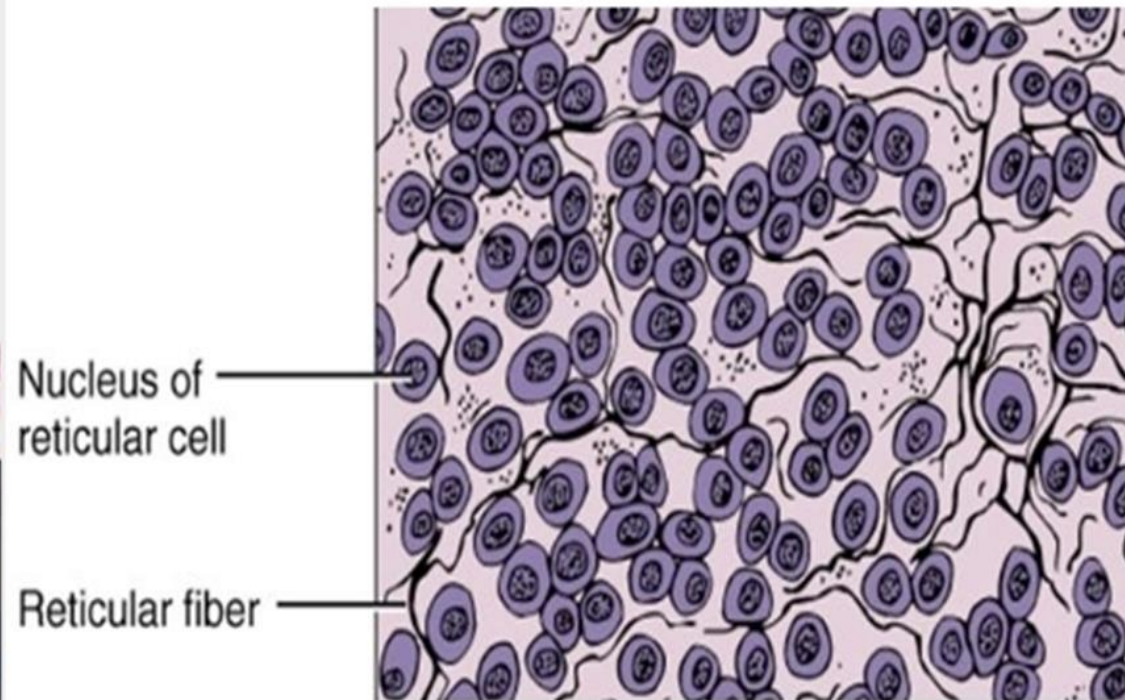
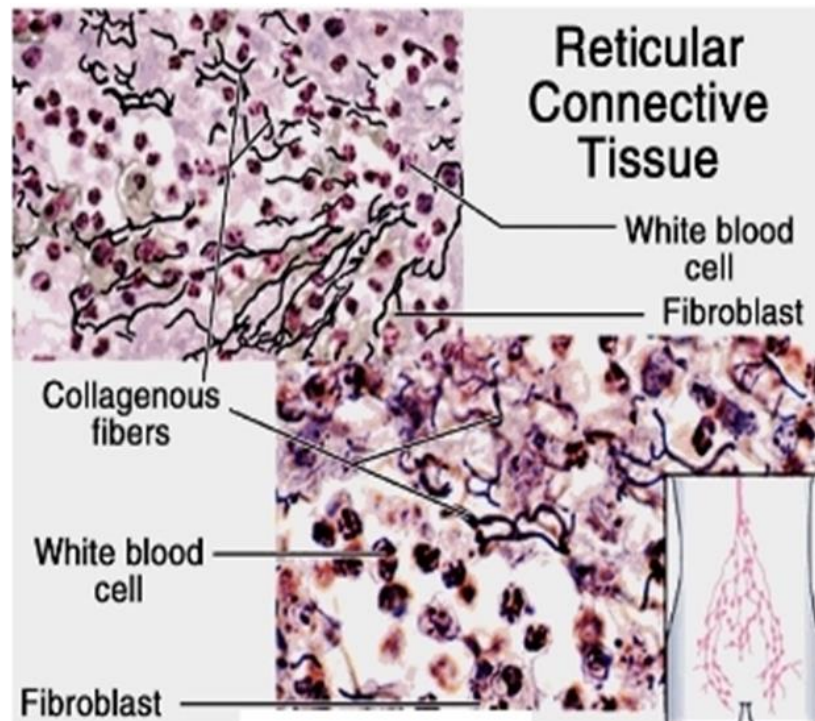
Special Connective Tissue

- **Special connective tissue includes:**

1. **Reticular Connective Tissue**
2. **Adipose Tissue (Body Fat)**
3. **Cartilage**
4. **Bones**
5. **Blood**

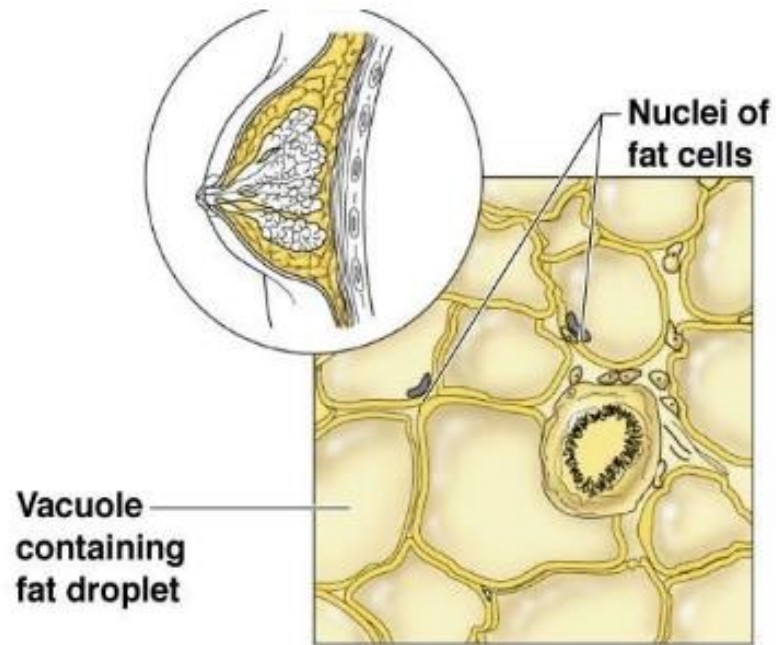
1. Reticular Connective Tissue

- **Extracellular matrix:** network of thin branched reticular fibers made of collagen called **reticular fibers**
- **Reticular fibers** are synthesized by special fibroblasts called **reticular cells**
- **Function:** support the lymphoid organs (red bone marrow, spleen) and hold adipose tissue in place by form the soft skeleton

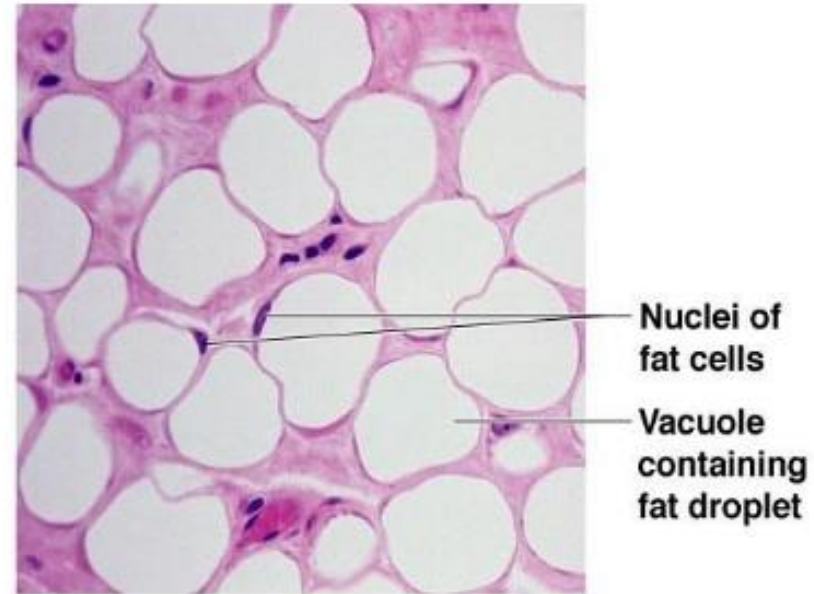


2. Adipose Connective Tissue

- Composed mostly of adipocytes, fibroblasts, vascular endothelial cells and adipose tissue macrophages.
- **Two types of adipose tissues :**
 - ✓ White adipose tissue (WAT), which stores energy
 - ✓ Brown adipose tissue, which generates body heat
- **Functions:**
 1. Store energy in the form of lipids
 2. Cushions
 3. Insulation of the body
 4. Production hormones such as leptin
- **Location: In humans, adipose tissue is located:**
 1. Beneath the skin (subcutaneous fat)
 2. Around internal organs (visceral fat)
 3. In bone marrow (yellow bone marrow)
 4. Intermuscular (Muscular system)
 5. In the breast tissue

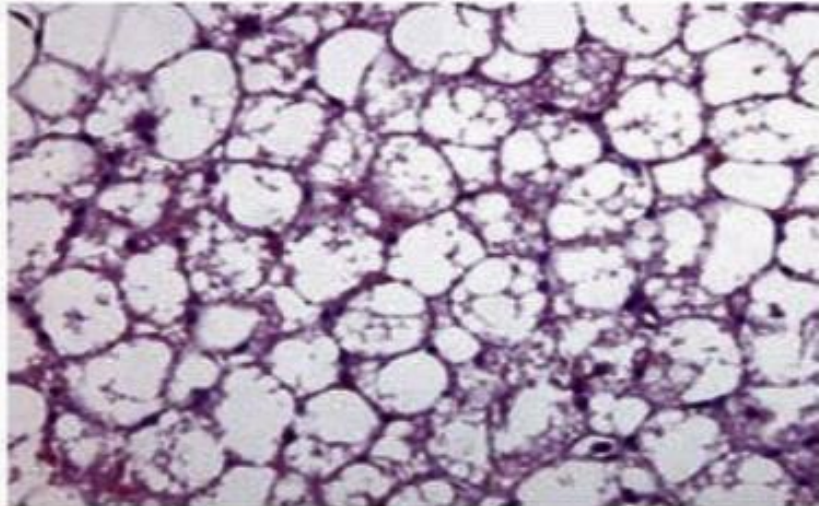


(f) Diagram: Adipose

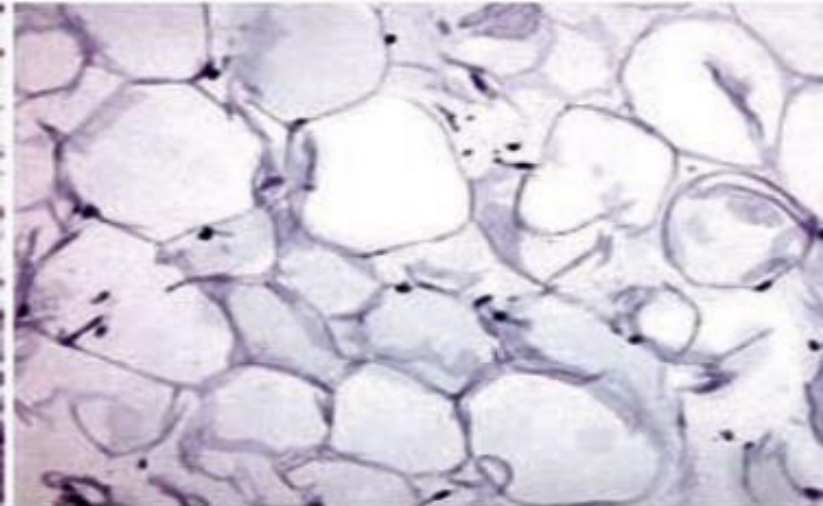


Photomicrograph: Adipose tissue from the subcutaneous layer beneath the skin (330x).

Brown Fat (or BAT)

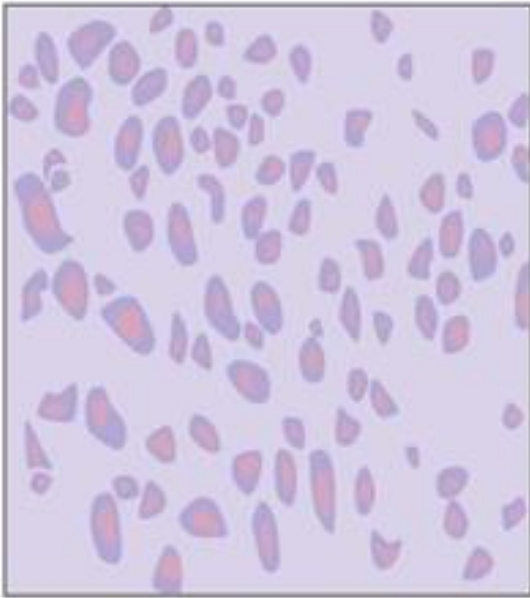


White Fat

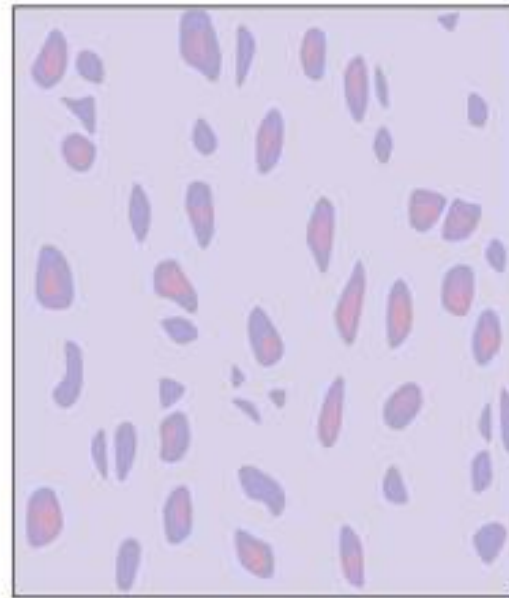


3. Cartilage

- Flexible connective tissue
- **Cell types:** chondrocytes that produce a large amount of collagenous extracellular matrix
- The cells found in a cavity called **lacuna**
- **Function:** gives shape, support, and structure to other body tissues.
- **Location:** in joints between bones, ribcage, ear, nose, bronchial tubes, intervertebral discs
- **There are three types of cartilage:**
 1. Elastic cartilage
 2. Hyaline cartilage
 3. Fibrocartilage



A. Elastic Cartilage

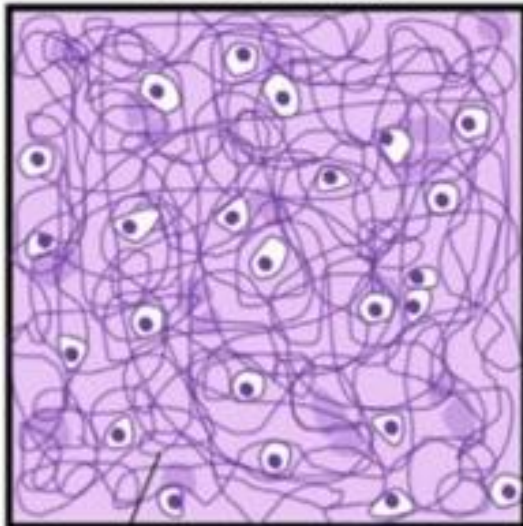


B. Hyaline Cartilage



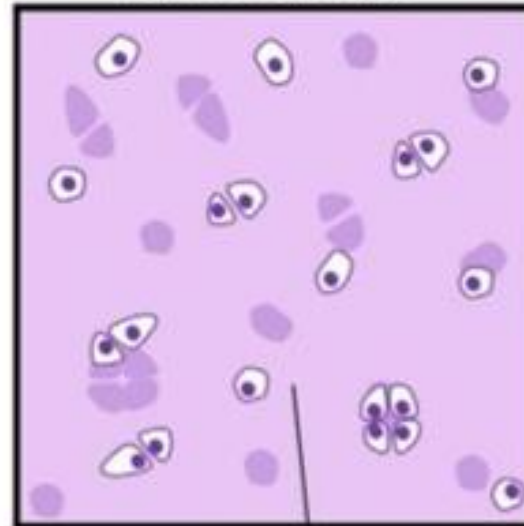
C. Fibrous Cartilage

Elastic cartilage



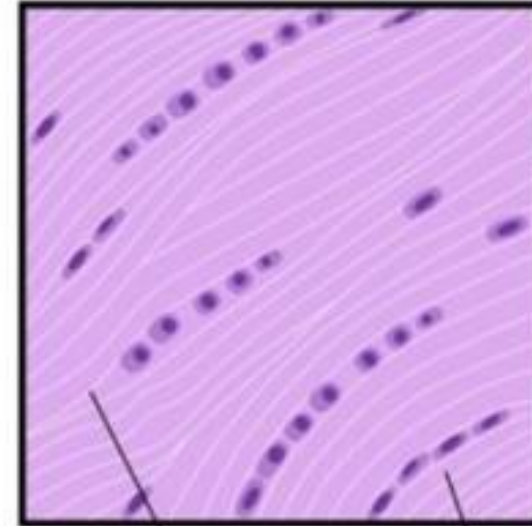
Visible elastic fibers in matrix

Hyaline cartilage



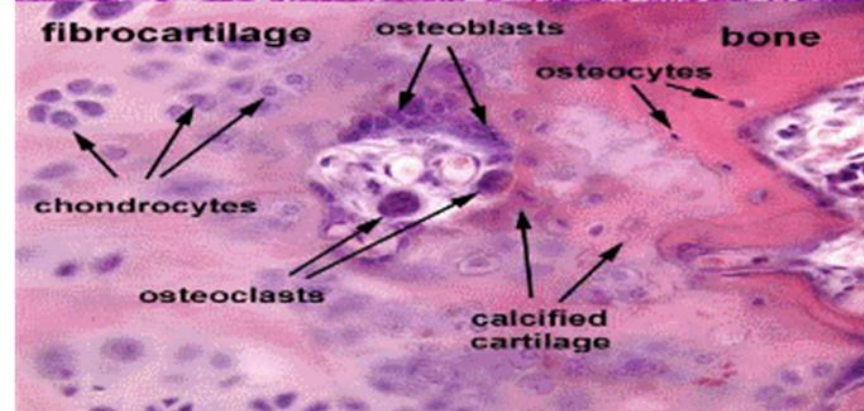
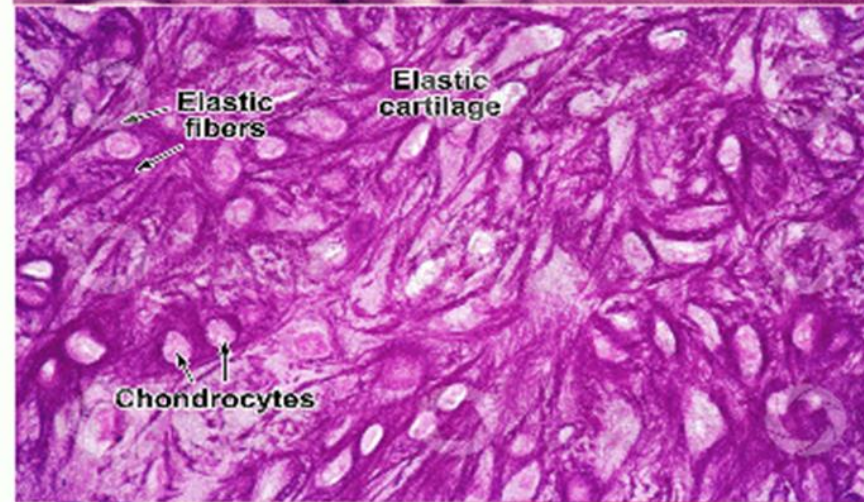
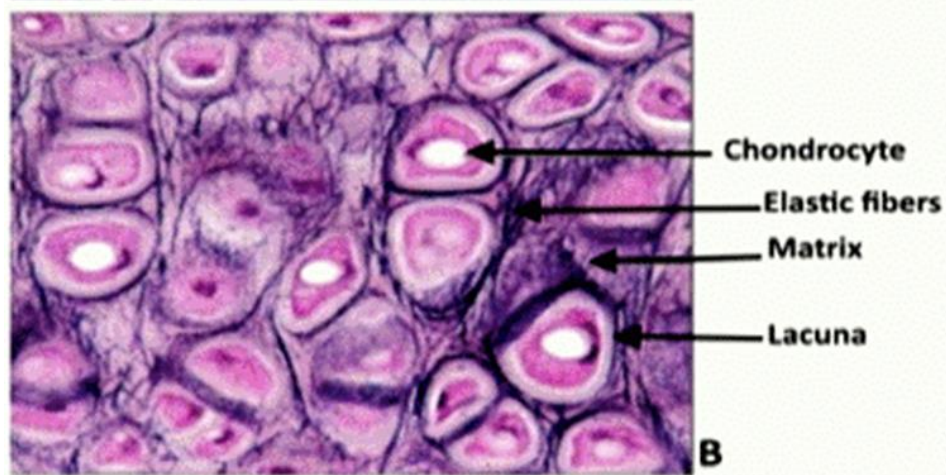
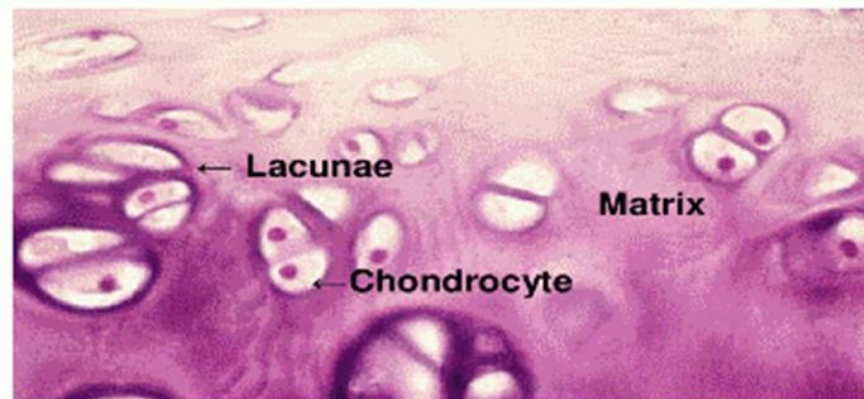
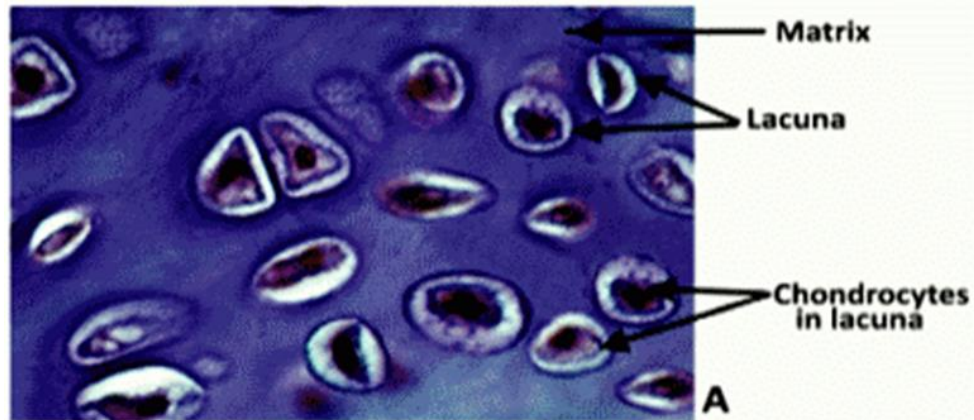
Clearer looking
ground substance

Fibrocartilage



Densely layered
collagen fibers

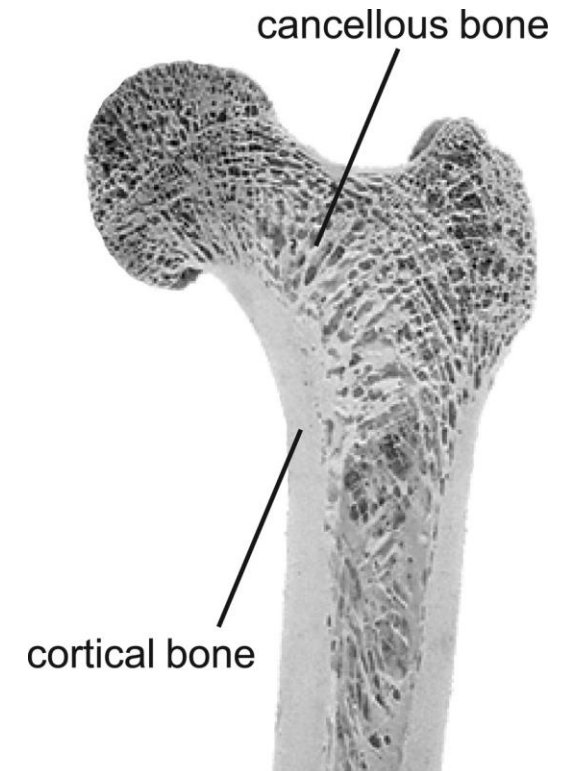
More flattened and
organized cell rows



Cartilage tissue: A) Hyaline; B) Elastic
C) Fibrocartilage

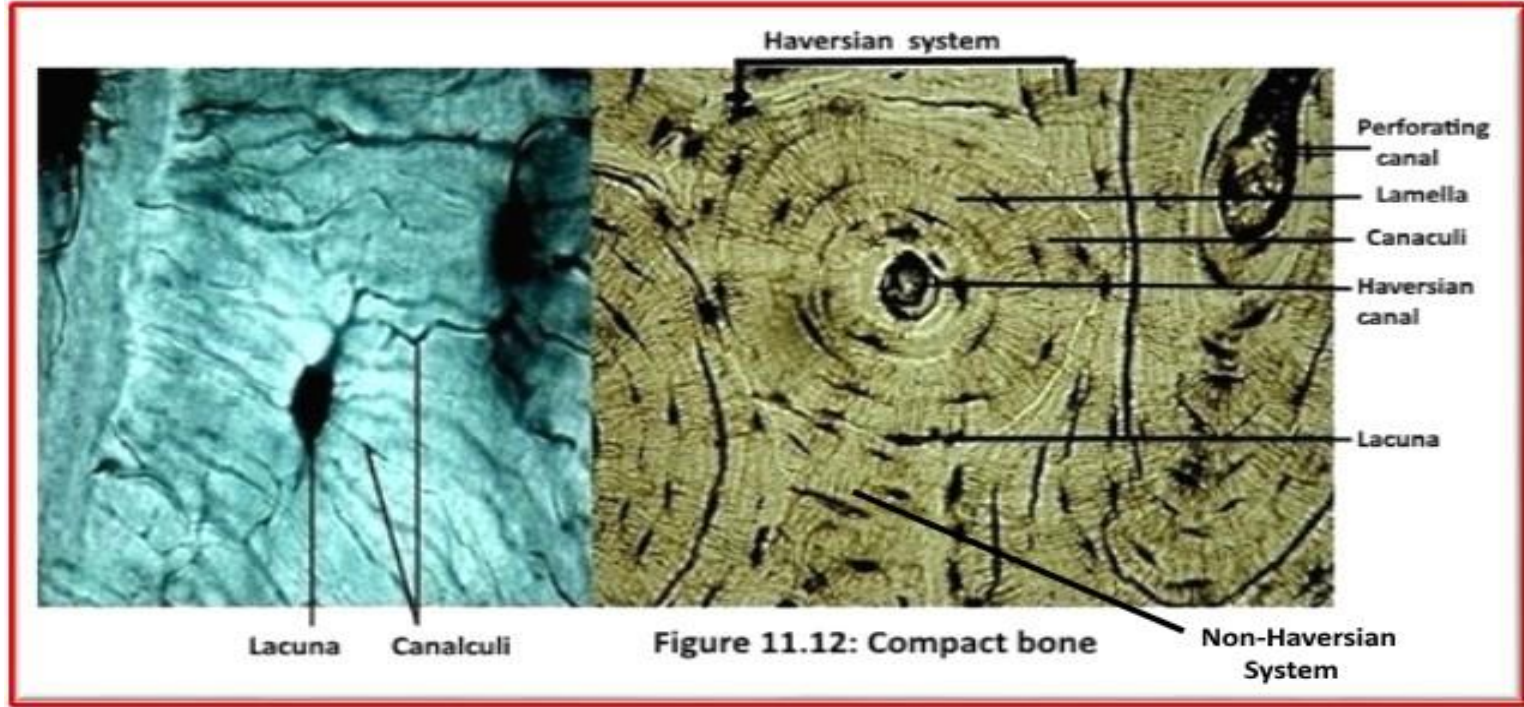
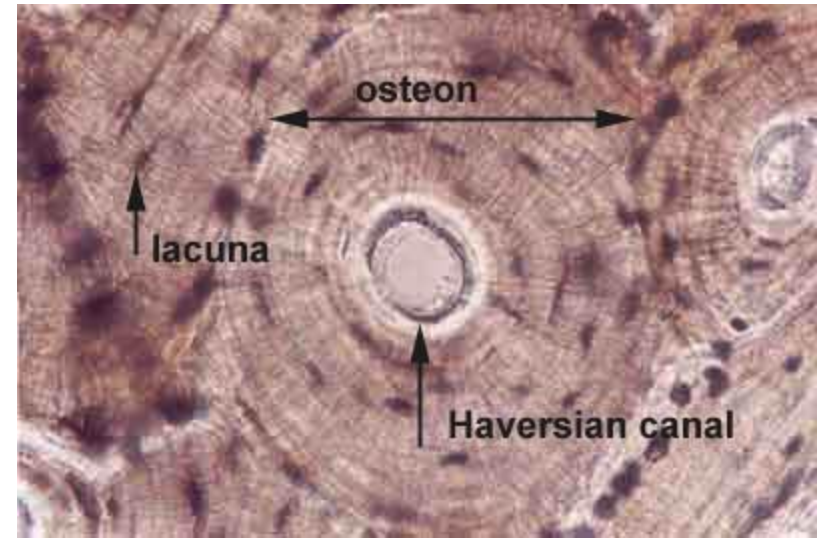
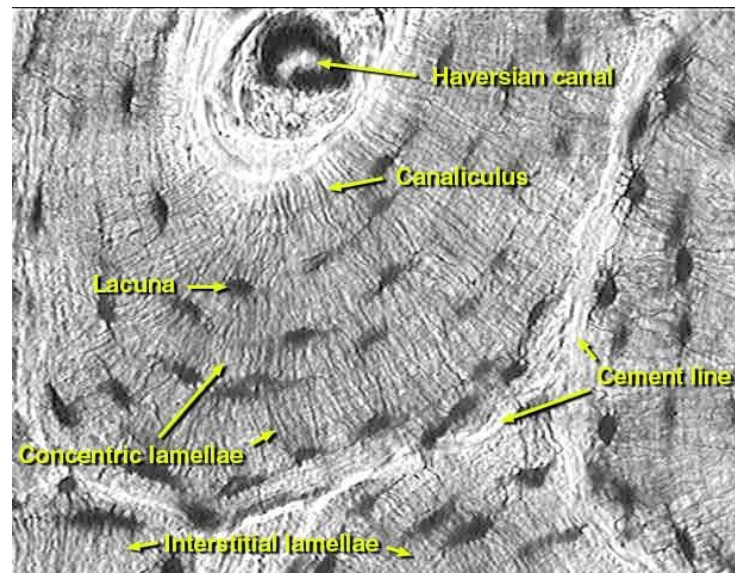
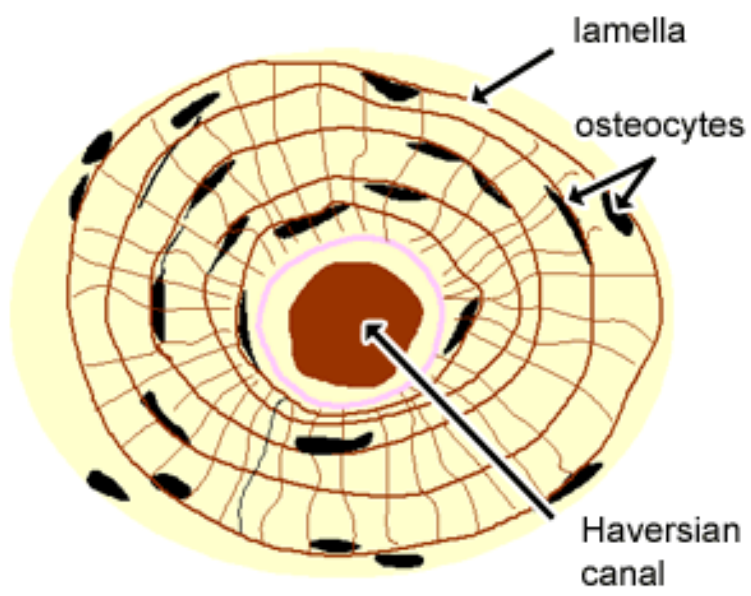
4. Bone

- **Location:** skeleton
- **Cell type:** osteocytes (living cells embedded in a mineralized organic matrix)
- **Extracellular matrix:** consists of organic components. The inorganic composition of bone (bone mineral) is primarily formed from salts of calcium and phosphate
- **Function:** support and protection
- There are two types of bone: **cortical bone (compact bone)** and **cancellous bone (spongy bone)**



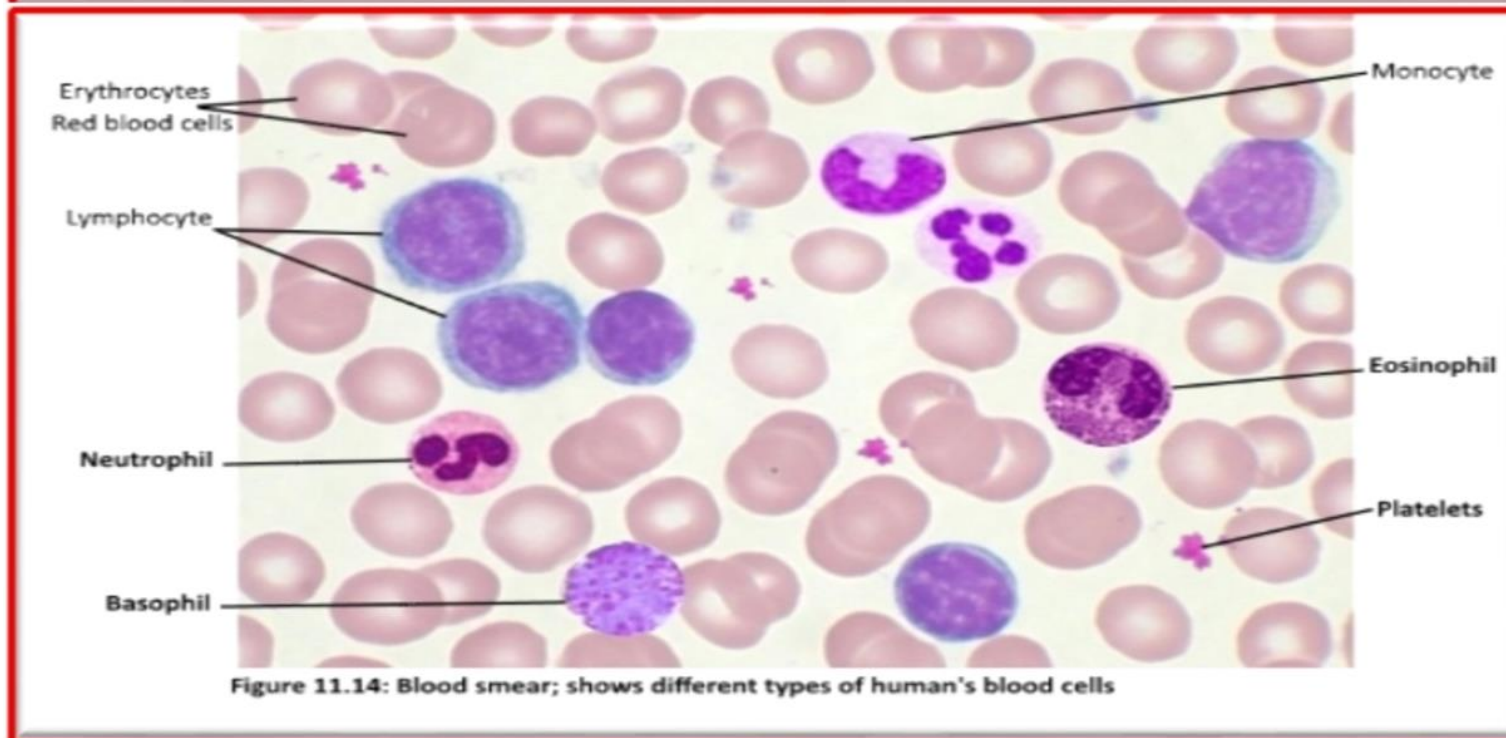
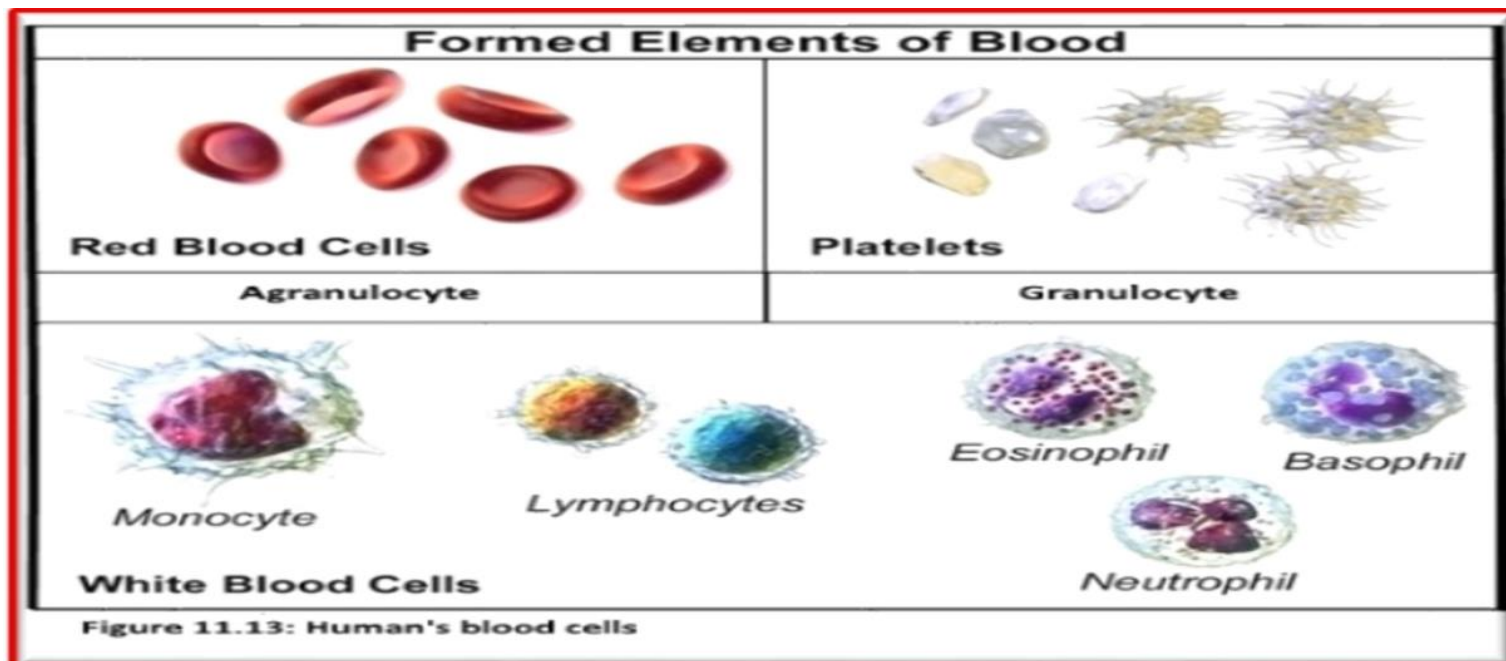
Cortical bone (Compact bone)

- **Osteons** are the primary anatomical and functional unit of cortical bone
- Cortical bone consists of a repeating structure called an **osteon**
- Each osteon consists of concentric **lamellae** (layers) of mineralized matrix
- **Lamellae** are deposited around a central canal, known as the **Haversian canal**
- Each osteon has its own blood and nerve supply
- **Lacunae** contain the bone cells (**Osteocytes**)



4. Blood

- **Function:** delivers nutrients and oxygen to the cells and transports metabolic waste products away from cells
- **Extracellular matrix:** in human composed of blood cells suspended in blood plasma (Extracellular matrix)
 - ✓ Blood plasma makes up 55% of blood, and it is 92% water by volume
- **Location:** blood vessels
- The blood cells are mainly:
 1. Red blood cells (RBCs or erythrocytes)
 2. Thrombocytes (platelets)
 3. White blood cells (also called WBCs or leukocytes)



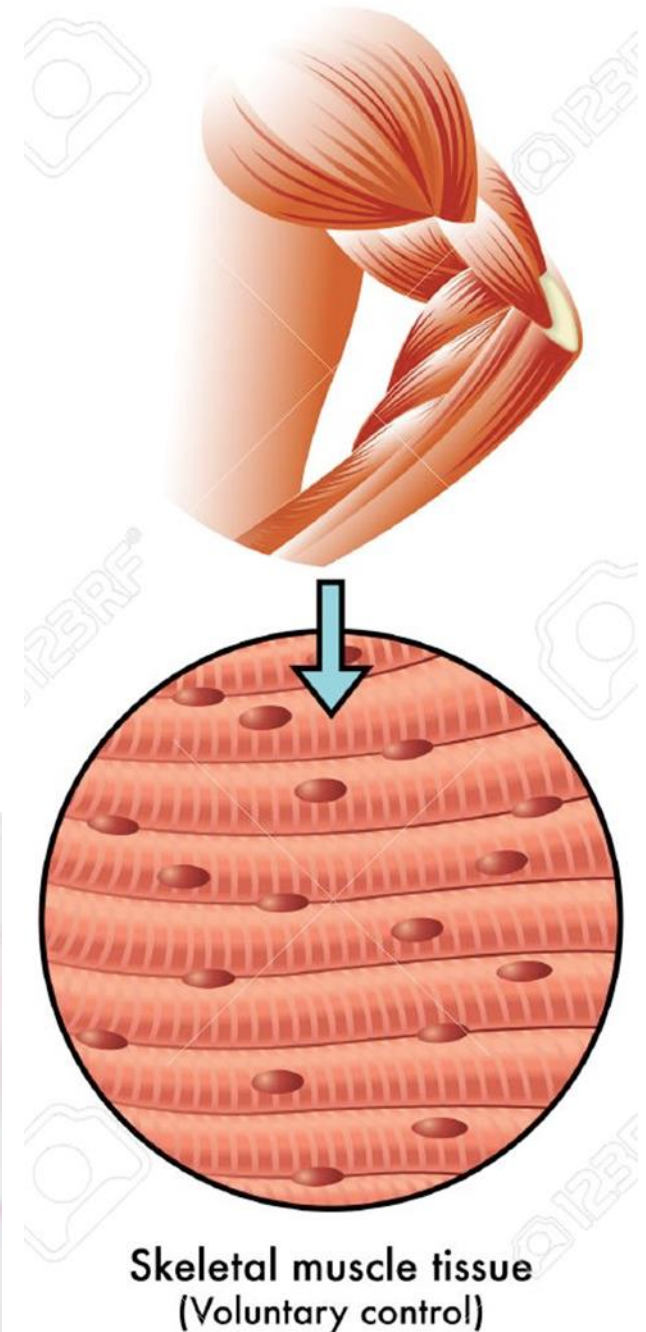
Muscle Tissues

Muscle Tissue

- **Function:** produce force and motion
- **Location:** anchored by tendons to bone (skeletal muscle); walls of organs and internal structures (smooth muscle);and heart (cardiac muscle)
- **There are three types of muscle tissue in mammals:**
 1. Skeletal muscle
 2. Smooth muscle
 3. Cardiac muscle (myocardium)

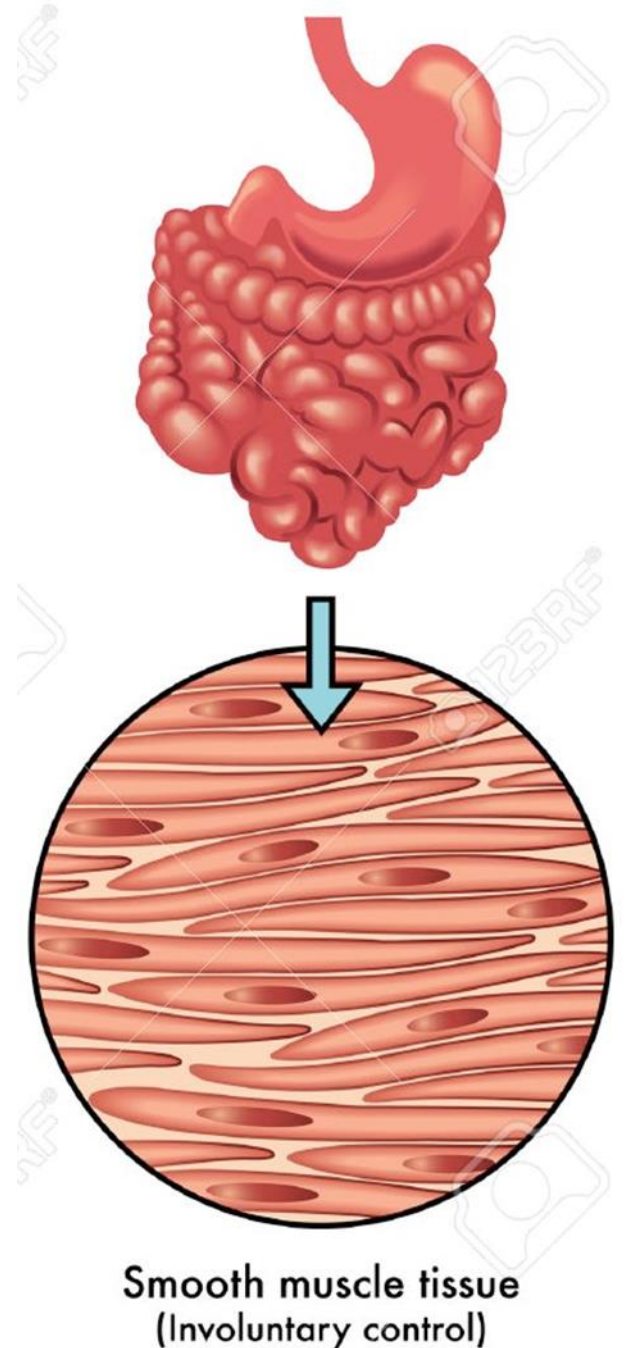
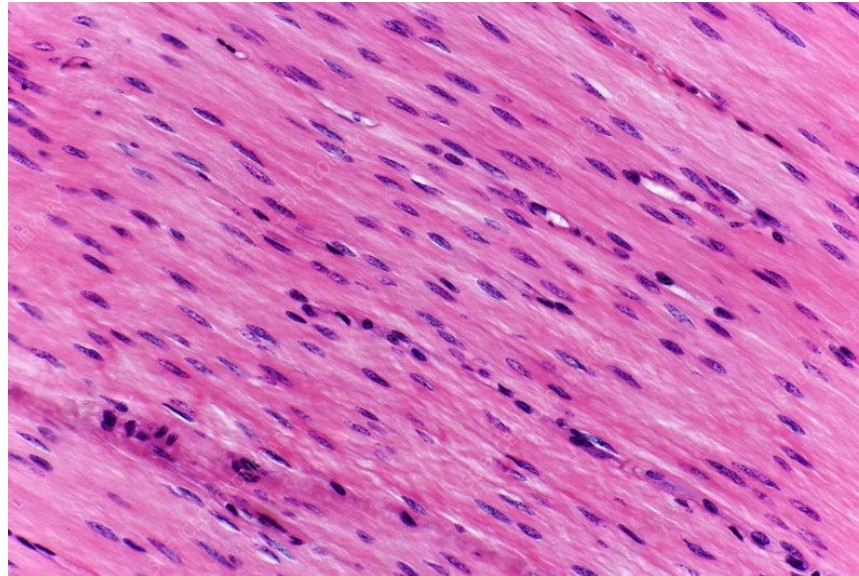
1. Skeletal Muscle Tissue

- **Function:** to effect skeletal movement such as locomotion
- **Location:** anchored by tendons to bones
- Striated
- Under voluntary control
- Males: 42% of body weight
- Females: 36% of body weight



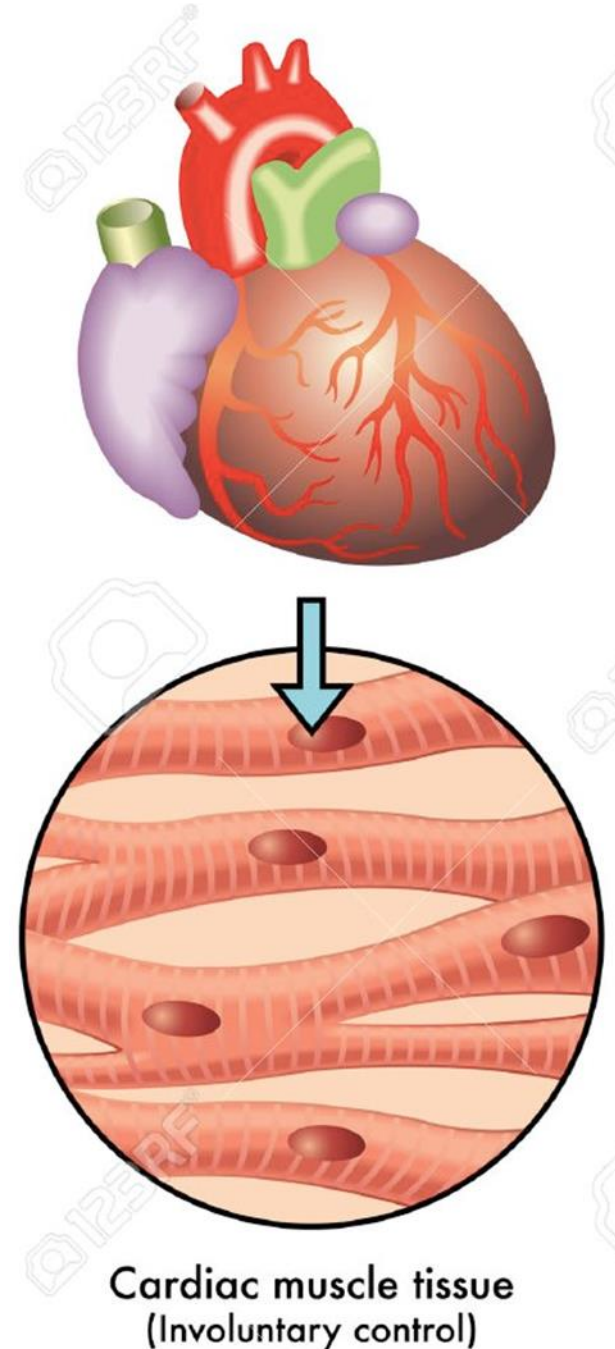
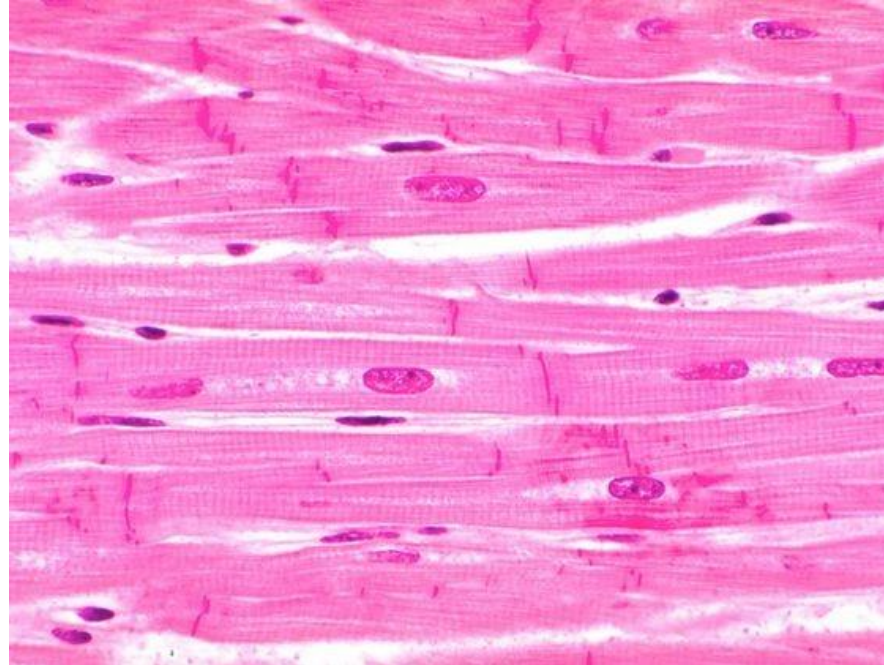
2. Smooth Muscle Tissue

- **Function:** to facilitate body functions
- **Location:** walls of organs and structures like the esophagus, stomach, intestines, bronchi, urethra, uterus, bladder, blood vessels, and arrector pili of skin
- Not striated
- Not under voluntary control
- **Activated by:** interaction of CNS



3. Cardiac Muscle Tissue

- Also known as myocardium
- **Function:** pump blood
- **Location:** heart
- Striated
- Not under voluntary control



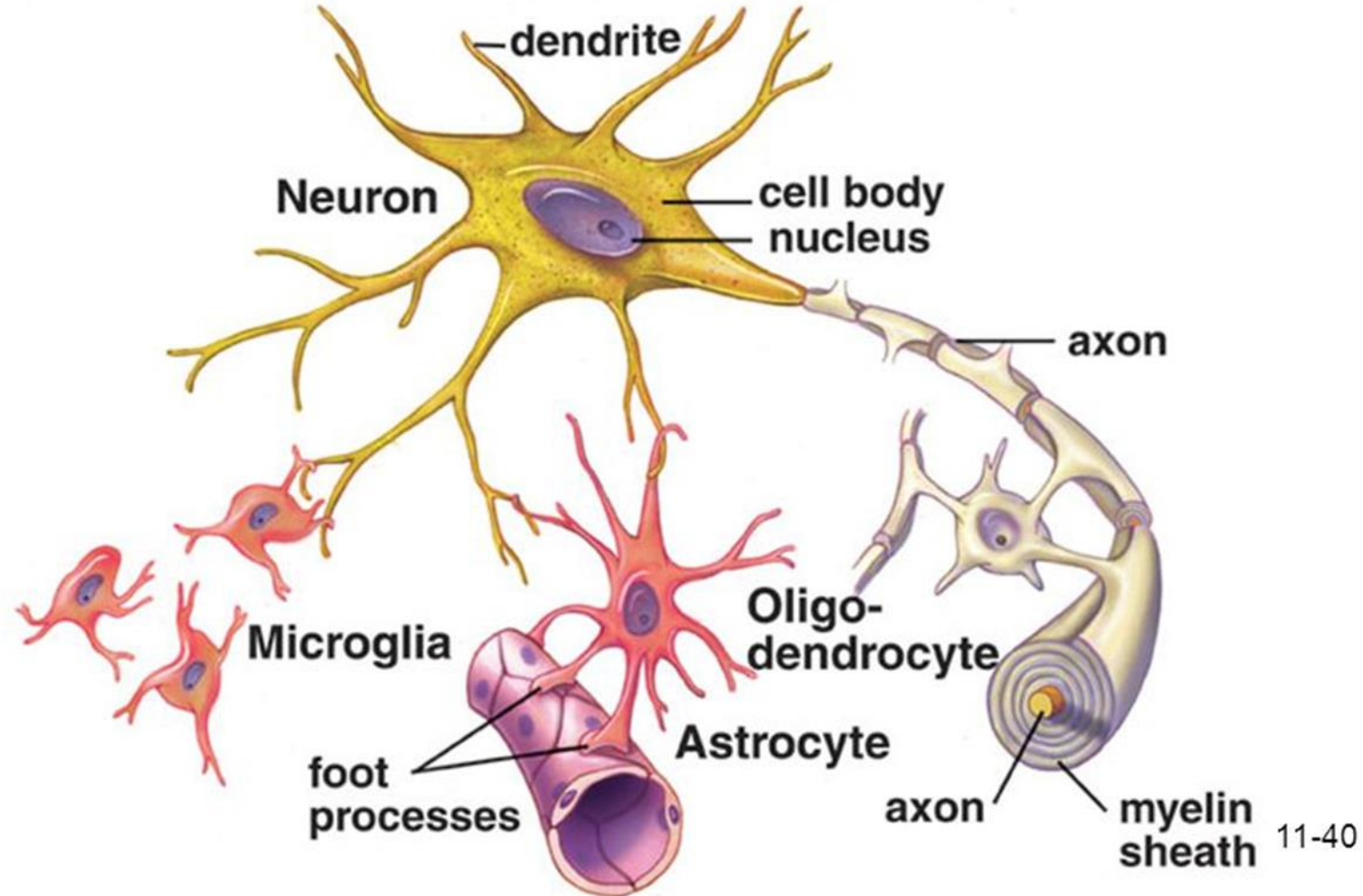
Nervous Tissue

Nervous Tissue

- **Function:** sensory input, integration, control of muscles and glands, homeostasis, mental activity
- **Location:** main component of the brain and spinal cord (CNS) and the nerves (PNS)
- The nervous tissue composed of two types of cells:
 1. **Neurons (nerve cells)**
 - Neurons receive and transmit impulses
 2. **Neuroglia (glial cells)**
 - Neuroglia assist the propagation of the nerve impulse and provide nutrients to the neuron

A neuron and some types of neuroglia

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



- **There are four types of nervous tissue:**

1. **Gray matter**

- ✓ Found in CNS
- ✓ Composed of cell bodies, dendrites, unmyelinated axons, protoplasmic astrocytes, microglia, and very few myelinated axons

2. **White matter**

- ✓ Found in CNS
- ✓ Composed of myelinated axons, fibrous astrocytes, myelinating oligodendrocytes, and microglia

3. **Nerves**

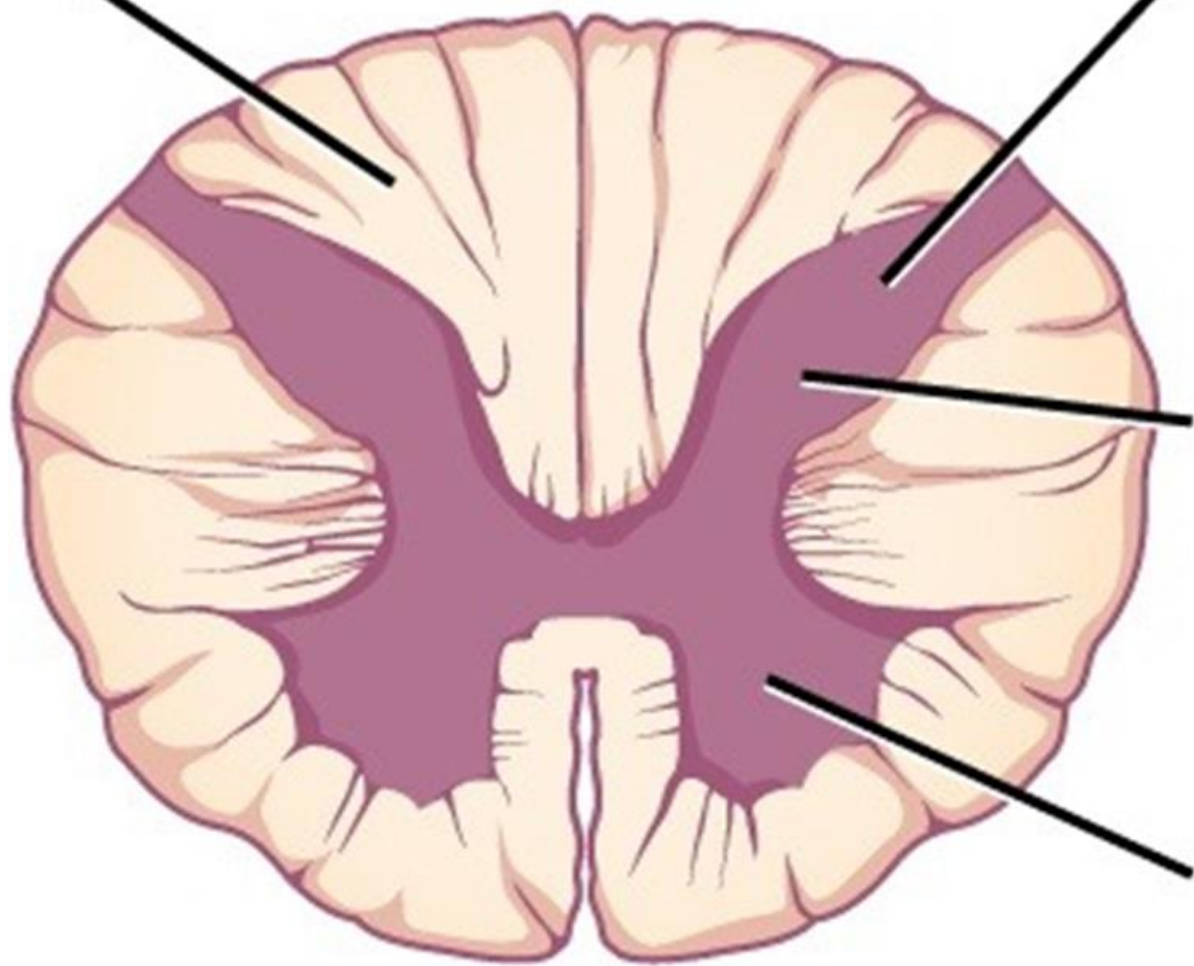
- ✓ Found in PNS

4. **Ganglia**

- ✓ Found in PNS

White matter

Gray matter



Dorsal horn

Ventral horn

